

Psychiatry AIIMS 2017

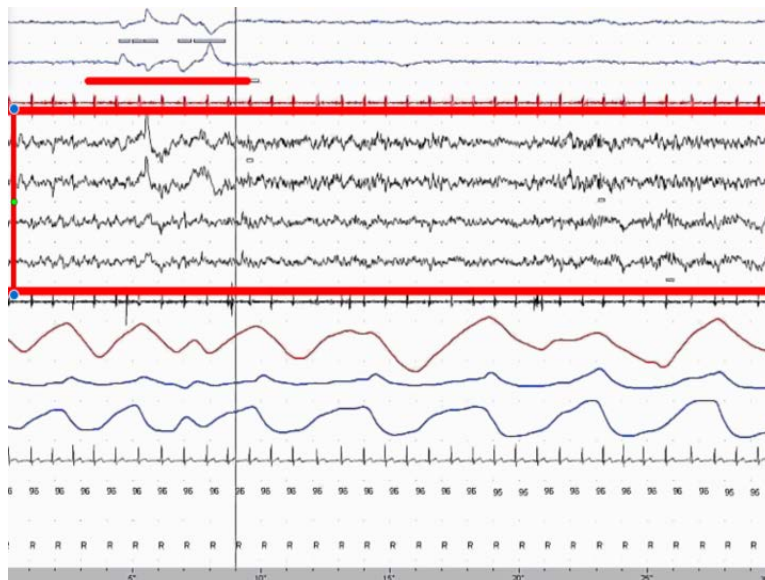
Question 1:

A 42-year-old man had a fall followed by one episode of vomiting. When brought to the casualty, he was awake and conscious but had unrelieved anterograde amnesia. Which of the following is the most likely cause?

- a) Drunkenness
- b) Diffuse axonal injury
- c) Concussion
- d) Cerebral venous thrombosis

Question 2:

The marked area given in the sleep polysomnography is of _____



- a) REM sleep
- b) NREM stage 1
- c) NREM stage 2
- d) NREM stage 3

Question 3:

Which of the following is the test for immediate memory?

- a) Digit span forward up to 7 digits with 2 skips allowed
- b) Serial (100-7) subtraction test up to 5 steps
- c) Digit span backwards up to 5 digits with 2 skips allowed
- d) Serial (20-1) subtraction test up to 5 steps

Question 4:

A patient believes he is the most important person in the world and so his neighbours and family members are trying to harm him as they are jealous of him. His wife says otherwise and says he behaves like this only recently and before that he was working peacefully as a school teacher. He is most likely suffering from?

- a) Delusion of grandiosity
- b) Delusion of persecution
- c) Delusion of grandiosity and persecution
- d) Delusion of grandiosity, persecution and reference

Question 5:

A young patient is admitted to hospital with acute psychosis. He wakes up and asks for his wife even while she is there in the same room right next to him. Upon pointing her out, he states that this is not his wife and someone else is masquerading as his wife. The most likely diagnosis is?

- a) Capgras syndrome
- b) Fregoli syndrome
- c) Delusion of subjective doubles
- d) Othello syndrome

Question 6:

A young male patient is on 5 mg haloperidol for many days. However, recently for the last 4 days, he has an inner restlessness and urges to move. The most likely diagnosis is:

- a) Akathisia
- b) Tardive dyskinesia

- c) Rabbit syndrome
- d) Acute dystonia

Question 7:

An IT employee who has recently taken up the job is feeling guilty, hopeless and not able to concentrate on work. His symptoms started 3 years back when he started college. The most likely diagnosis is:

- a) Depressive disorder
- b) Dysthymia
- c) Adjustment disorder
- d) Cyclothymia

Question 8:

Mutism and akinesia in a person who is aware of his surroundings and somewhat alert is best described as a feature of:

- a) Oneroid state
- b) Twilight state
- c) Stupor
- d) Delirium

Question 9:

A 25 year old female has thoughts of cutting her fingers, she plans and imagines doing it but never actually does it. She says she is not having any guilt of having such thought. And also says the thoughts are distressing her and she is unable to control them. The thoughts vanish either by ending with a seizure or automatically subsides on its own. Which of the following is the cause?

- a) Thought insertion
- b) Forced thinking
- c) Crowding of thoughts
- d) Obsession

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	c
5	a
6	a
7	b
8	c
9	b

Detailed Explanations

Solution to Question 1:

The given history of fall and anterograde amnesia points towards the diagnosis of concussion.

A concussion is the most common type of traumatic brain injury. It can result from vehicle accidents, falls, sports injuries, etc. Anterograde amnesia/post-traumatic amnesia is the hallmark of a concussion. In this type of amnesia, the events following the injury cannot be recalled.

Other clinical features of concussion include headache, dizziness, nausea, and vomiting.

Option A: Drunkenness - the patient will be drowsy or sleepy and not awake.

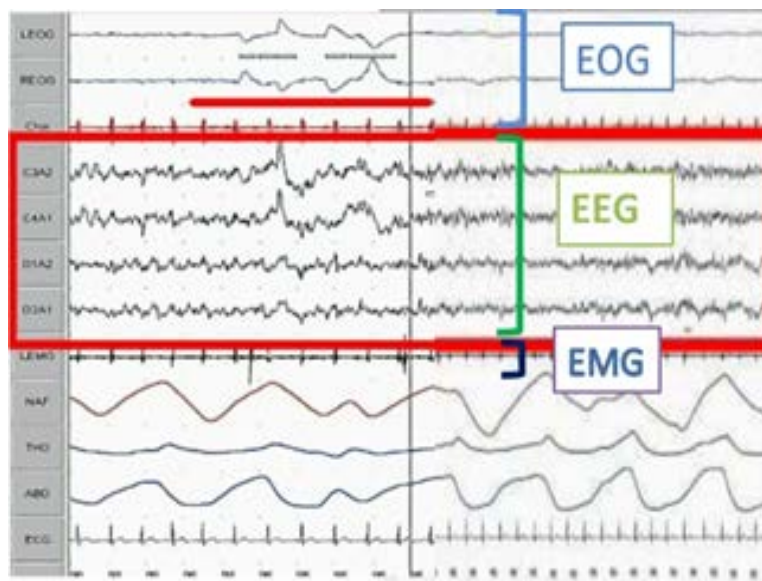
Option B: Diffuse axonal injury - seen in high-speed deceleration which causes disruption and shearing of axons leading to microhemorrhages. Usually results in a persistent vegetative stage.

Option D: Cerebral venous thrombosis - acute thrombosis in dural venous sinuses; manifests as headache, visual abnormalities, or seizures.

Solution to Question 2:

The marked area in the given polysomnography represents REM sleep.

The three zones in the given image represent EOG, EEG and EMG, respectively.



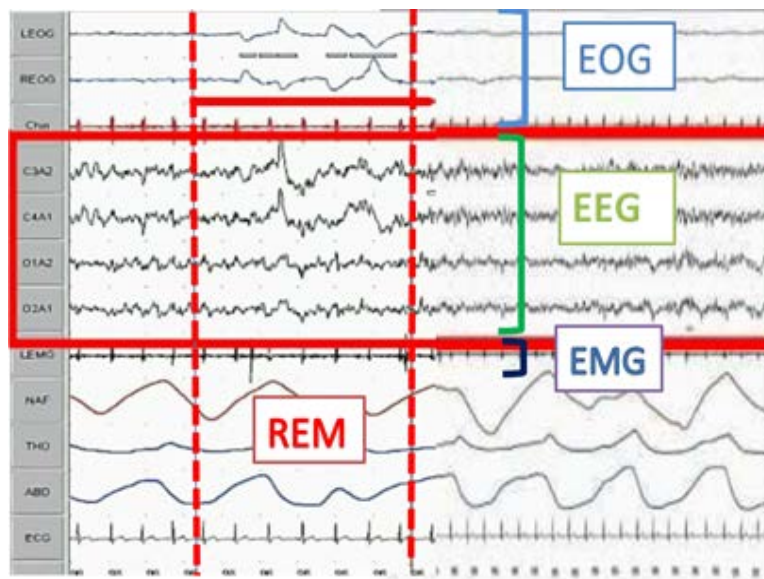
The following table helps identify the state of wakefulness from the graph :

In the marked area,

- EOG shows Positive activity
- EEG shows Mixed waves
- EMG shows Minimal activity

This is characteristic of REM sleep.

	EEG	EOG	EMG
Active state	Fast	+	+
REM sleep	Mixed	+	-
NREM sleep	Slow	-	+



Solution to Question 3:

Forward digit span test is the best test among the given options.

Memory is of 3 types:

- Immediate
- Recent
- Remote

Immediate memory is attention dependent. Tests of attention and memory include

- Digit span test: Most important and best test for immediate memory
- Serial (100-7) subtraction test is the best for concentration
- Days of the week backwards
- Spell WORLD backwards
- (20-3) test

Digit span test

The examiner recites numbers randomly without any sequence and asks the patient to repeat the same. The examiner starts with a 2-number sequence and the digit span increases from 2 numbers to 3 numbers and so on. The patient is asked to repeat until he/she fails to repeat all numbers correctly.

Digit forward

- Normally a person can repeat up to 7 digits (+/- 2)
- E.g., Examiner says 2 - 4 - 8 - 1 - 5 - 7 - 9. The person has to repeat in the same order 2 - 4 - 8 - 1 - 5 - 7 - 9.

Digit backward

- Reproduction of longer digits in backward direction.
- Normally a person can repeat up to 5 digits (+/- 1) in backward direction (making option C wrong)
- Eg : Examiner says 2 - 4 - 8 - 1 - 5 - 7 - 9. The person has to repeat it backwards. 9 - 7 - 5 - 1 - 8 - 4 - 2.

Serial (100-7) subtraction test

- 5 scores / steps are taken in to account
- 1st score – When the patient tells 93 ($100 - 7 = 93$)
- 2nd score - When the patient tells 86 ($93 - 7 = 86$)
- And so on . . 100 , 93 , 86 , 79 , 72 , 65

Hence among the given options :

Digit span forward up to 7 digits with 2 skips allowed would be the answer of choice as it is the best test for immediate memory

Solution to Question 4:

- Belief that he is the most important person in the world – Delusion of grandeur
- Belief that his neighbours and family members are trying to harm him – Delusion of persecution
- Belief that others feel jealous – Part of delusion of persecution

A scenario of this kind is most likely seen in manic patients with

- Primary delusion of Grandeur (and)
- Secondary delusion of Persecution

Delusion of reference:

- Delusion in which events, objects, or other persons in one's immediate environment have a particular and unusual significance in reference to the person experiencing the delusion.
- Examples :
- A feeling that people on television or radio are talking about or talking directly to them
- Believing that headlines or stories in newspapers are written especially for them

Solution to Question 5:

According to the given scenario,

A stranger has taken the guise of Wife (Family person)

This is characteristic of Capgras syndrome.

Capgras syndrome

The delusion in Capgras syndrome is the belief that a familiar person has been replaced by an imposter

Fregoli's phenomenon

- Fregoli syndrome is a disorder in which a person holds a delusional belief that different people are in fact a single person who changes his or her appearance or is in disguise.

Intermetamorphosis

- A very rare delusion that familiar persons can change themselves into other persons at will.

Doppelganger

- A person feels he has his own impostor (duplicate) – Delusion of subjective doubles

Othello syndrome

- It is a rare psychiatric condition marked by morbid, pathological, or delusional jealousy.
- Eg : Suspecting a faithful partner of infidelity, with accompanying jealousy.

Solution to Question 6:

The probable diagnosis is Akathisia.

Haloperidol is an antipsychotic drug and a side effect is Extrapyramidal symptoms.

Inner restlessness and urge to move are EPS referred to as Akathisia.

Akathisia

Akathisia consists of motor restlessness accompanied by subjective feelings of inner tension and discomfort referable chiefly to the limbs.

Restless leg syndrome

Closely resembles akathisia but occurs in absence of neuroleptic medication. No neurological deficit is seen.

Tardive Dyskinesia

Tardive dyskinesia is involuntary movements of the tongue, lips, face, trunk, and extremities that occur in patients treated with long-term dopaminergic antagonist medications.

Rabbit syndrome

Rabbit syndrome is an antipsychotic-induced rhythmic motion of the mouth/lips, resembling the chewing movements of a rabbit.

The movement consists of a vertical-only motion, at about 5Hz, with no involvement of the tongue.

Solution to Question 7:

The above scenario is consistent with the diagnosis of dysthymia.

Dysthymia / Persistent Depressive Disorder:

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DSM V Criteria for diagnosis of dysthymia: Depressed mood for most of the day as indicated by either subjective account or observation by others, for at least 2 years. In children and adolescents - at least 1 year.

Presence of two (or more) of the following while being depressed.

- Poor appetite or overeating
- Insomnia or hypersomnia
- Low energy or fatigue
- Low self-esteem
- Poor concentration and/or difficulty making decisions
- Feelings of hopelessness

Early-onset, beginning in childhood or adolescence and in most cases by the age of 20 years. More common in unmarried and young persons with low income.

Cyclothymia:

- A milder form of bipolar disorder which is chronic in nature.
- Characterized by episodic mood swings of hypomania and mild depression.

Adjustment disorder:

- It is an abnormal and excessive reaction to an identifiable life stressor.
- It is a temporary condition
- Eg: Feeling sad or hopeless in an exaggerated manner after a stressful life event. Inability to cope with stressful situations.

Solution to Question 8:

Akinetic mutism is a feature of Stupor.

- In this condition, the patient is unable to initiate action and speech but appears awake and alert.
- There is severe reduction of relational function – speech and action.

Akinetic mutism is also seen in Schizophrenia with affective psychosis and dissociative states.

Stupor :

- Lesion at :
- Diencephalon
- Upper brain stem
- Frontal lobes
- Basal ganglia

- Locked In syndrome

Option A, Oneroid state

- Impaired consciousness.
- Marked emotional changes in response to visual hallucination.
- Patient lives as if in a dream state – occupational delirium.

Option B, Twilight state

- Abrupt onset of abnormal violent behavior
- Occurs in
- Epilepsy
- Alcoholism
- Brain trauma

Option D, Delirium

- Global impairment of consciousness and cognition.

Solution to Question 9:

In the above scenario, the patient has forced thinking.

Forced thinking is usually seen in patients with frontal lobe epilepsy. The thoughts are formed are not in the control of the patient and are distressing thoughts.

Option A: thought insertion is seen in schizophrenia. The delusion of thoughts being placed into patient's mind by someone else.

Option C: crowding of thoughts is also seen in schizophrenia. The patient has thoughts that are too fast and excessive and unexplainable. The patient can even locate these thoughts crowding at the back of the head.

Option D: The obsessive-compulsive symptoms are not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication) or another medical condition. The patient feels the need to carry out certain behaviours to get rid of these thoughts.

Psychiatry AIIMS 2018

Question 1:

A patient was prescribed escitalopram. Which of the following adverse effect is the least likely?

- a) Nausea
- b) Vivid dreams
- c) Anorgasmia
- d) Sialorrhea

Question 2:

Abrupt stoppage of which of the following drugs causes agitation, anxiety and insomnia?

- a) Valporate
- b) Olanzapine
- c) Imipramine
- d) Venlafaxine

Question 3:

A young male who is shy and prefers social isolation appears to be emotionally cold and resides alone. There is no history of hallucinations or delusions. The most likely diagnosis is

- a) Paranoid personality disorder
- b) Schizoid personality disorder
- c) Antisocial personality disorder
- d) Emotionally unstable personality disorder

Question 4:

A 14 year old girl presents to the emergency room with fracture forearm and says she has tripped and fell. On examination you notice multiple cigarette burns on her forearm. What

will you do next?

- a) Conduct a thorough examination
- b) Seek a social worker's help
- c) Inform higher authorities immediately
- d) Inform your colleagues

Question 5:

Among the following, what is the most common symptom of adult obsessive-compulsive disorder?

- a) Aggressive symptoms
- b) Need for symmetry
- c) Pathological doubt
- d) Sexual symptoms

Question 6:

Which of the following is not a feature of atypical depression?

- a) Increased sleep
- b) Reactive mood response to positive stimulus
- c) Heaviness of the body
- d) Responds better to TCAs than MAO-I and SSRI

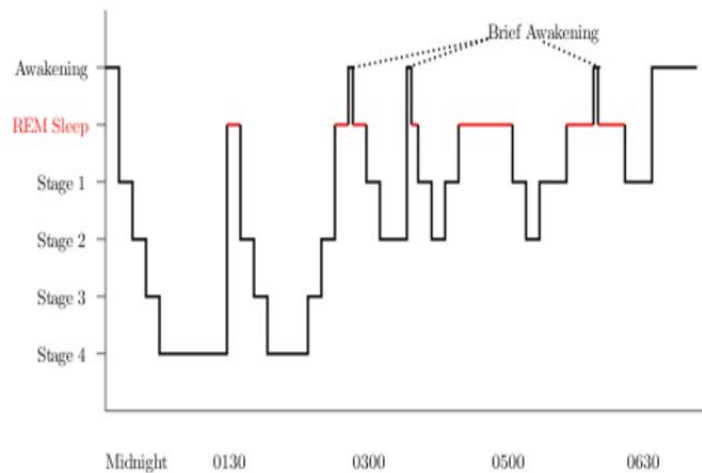
Question 7:

Under the Mental Health Care Act 2017, a patient has the right to choose his or her caretaker and future course of action in treatment. What is this provision known as?

- a) Treatment directive
- b) Mental will
- c) Future directive
- d) Advance directive

Question 8:

What does the given image represent?



- a) Hypnogram
- b) Electroencephalogram
- c) Polysomnogram
- d) Epworth sleep assessment

Question 9:

A 40-year-old woman has a history of urinating in public without having any remorse for it. She also has difficulty in getting motivated and has mild memory deficits. Which of the following lobes may be affected?

- a) Parietal
- b) Occipital
- c) Frontal
- d) Temporal

Answer Key

Question No.	Correct Option
1	d

2	d
3	b
4	a
5	c
6	d
7	d
8	a
9	c

Detailed Explanations

Solution to Question 1:

Sialorrhea is not a side effect of escitalopram.

Escitalopram is a selective serotonin reuptake inhibitor (SSRI). SSRIs are associated with dry mouth, and not sialorrhea.

System	Side effects of SSRIs
Sexual	Anorgasmia, Decreased libido
Gastrointestinal (5-HT ₃ receptor)	Nausea, Diarrhea, Anorexia, Vomiting, Flatulence, and Dyspepsia
Nervous system	Extremely vivid dreams or nightmares, Anxiety, Insomnia/sedation, Seizures, Emotional blunting
Anticholinergic	Dry mouth, Constipation, Sedation
Cardiovascular	Drug induced long QT syndrome
Hematological	Easy bruising
Endocrine	Mammoplasia, Galactorrhea

Solution to Question 2:

Abrupt stoppage of venlafaxine causes agitation, anxiety, and insomnia.

Solution to Question 3:

The diagnosis, in this case, is schizoid personality disorder.

Schizoid personality disorder is a cluster A personality disorder. The patient neither desires nor enjoys close relationships (shy and prefers social isolation, resides alone as in the question) and shows emotional coldness and detachment.

In contrast, subjects with emotionally unstable (borderline) personality disorder cannot tolerate being alone, and they search for companionship.

Solution to Question 4:

A child who has been physically abused should undergo a thorough physical examination at first.

This should be followed by a psychiatric interview.

When the child is brought to the emergency room, a detailed and spontaneous account of the injury should be obtained promptly from parents or other caregivers before secondary details and rationalizations cloud the information provided.

Solution to Question 5:

Among the above options, pathological doubt is the common symptom of adult OCD. Pathological doubt is the second most common pattern of adult obsessive-compulsive disorder (OCD), only after obsession of contamination.

Pathological doubt is an obsession with doubt, followed by a compulsion to check. The obsession often implies some danger of violence e.g., forgetting to turn off the stove or not locking a door. The checking may involve multiple trips back into the house to check the stove. These patients have obsessional self-doubt and always feel guilty about having forgotten or committed something.

The frequency of presenting patterns of OCD :

Obsession of contamination > Pathological doubt > Intrusive thoughts> Symmetry

Solution to Question 6:

Patients with atypical features do respond effectively to treatment with monoamine oxidase inhibitors (MAOIs) and selective serotonin reuptake inhibitors (SSRIs).

Mood reactivity, increased sleep, weight gain, and leaden paralysis (heaviness of body) are features of atypical depression.

Mood reactivity is the capacity to be cheered up when presented with positive events.

Leaden paralysis is the heavy, leaden feelings in arms or legs.

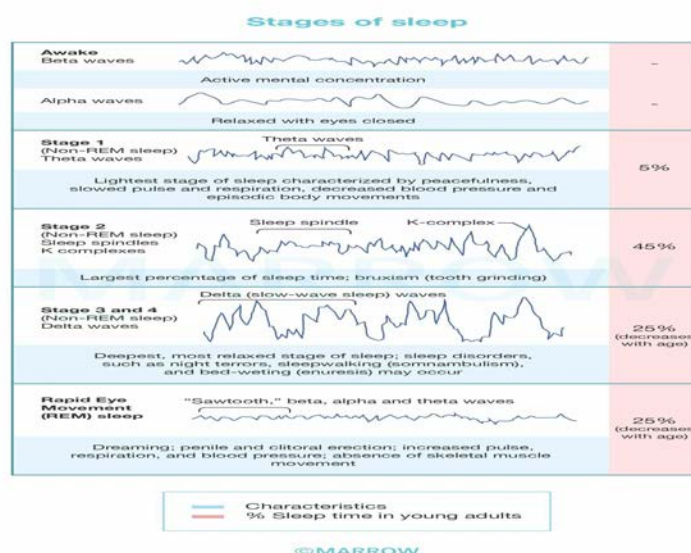
Solution to Question 7:

The provision by which a patient has the right to choose his caretaker and future course of action in treatment because of illness or incapacity is known as an advance directive. It is also known as a living will, personal directive, or medical directive.

Solution to Question 8:

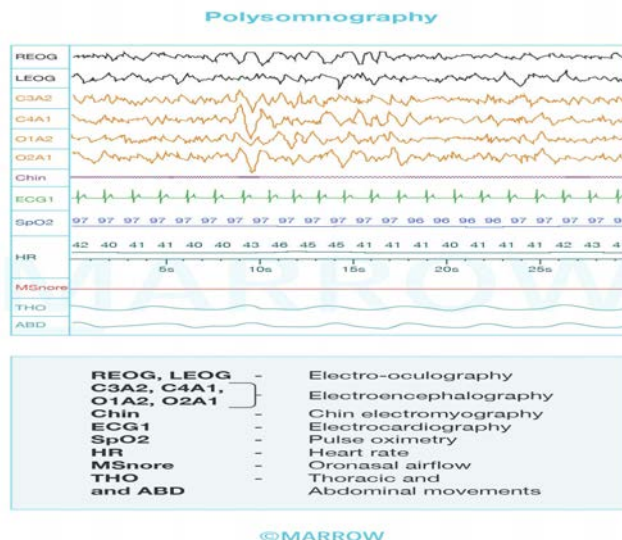
The above image represents a hypnogram. It is a graph that depicts the stages of sleep as a function of time. It provides a visual representation of sleep structure and is useful in the diagnosis of various sleep disorders.

An electroencephalogram is the depiction of the electrical activity of the surface of the brain. Different waveforms (alpha, beta, theta, delta) are produced depending on the functional status of the brain.



Polysomnography is a multi parametric test that is considered the gold standard in the diagnosis of sleep disorders. It is a continuous recording of biophysiological changes that occur during sleep. It includes,

- Continuous EEG recording, from occipital and parietal leads
- Electro-oculography- to record eye movements
- Electromyography- for muscle potential and its activities
- ECG
- Multiple sleep latency tests- repeated measures of time taken to sleep
- Nasal-oral airflow, respiratory effort, oxyhemoglobin saturation, and leg movements during sleep are measured.



Epworth sleep assessment scale is a questionnaire used to quantify daytime sleepiness. It determines how frequently a patient is likely to doze off in 8 commonly encountered situations. It is a subjective assessment and hence less reliable.

Solution to Question 9:

The frontal lobe is affected in this patient. The woman has difficulty in getting motivated, memory deficits, and urinary incontinence.

Frontal lobe injury is commonly due to trauma, infarcts, tumors, lobotomy, multiple sclerosis, or Pick's disease. The following features may be seen in a patient with frontal lobe injury:

- Impaired executive functions such as motivation, attention, and sequencing of actions
- Aphasia, apraxia, constructional apraxia, neglect, and memory deficits
- Urinary incontinence and fecal incontinence is commonly seen
- Changes in personality
- Apathetic indifference and impulsive behavior.

Note: Frontal lobe pathology may become apparent only under unstructured, stressful, real-life situations and are difficult to detect using Intelligence scales, as IQ mostly requires parietal lobe activation.

Psychiatry AIIMS 2019 (May & Nov)

Question 1:

Under the Mental Health Care Act 2017, a patient has the right to choose his or her caretaker and future course of action in treatment. What is this provision known as?

- a) Treatment directive
- b) Mental will
- c) Future directive
- d) Advance directive

Question 2:

Which of the following is not a feature of melancholia?

- a) Severe anhedonia
- b) Weight loss
- c) Sleep-onset insomnia
- d) Profound feelings of guilt over trivial events

Question 3:

Which of the following best describes paradoxical suicide?

- a) Suicide occurring at the time when the patient starts to recover
- b) Suicide after taking low dose of drug
- c) Accidental completion of suicide
- d) None of the above

Question 4:

Which of the following is true regarding depersonalization disorder?

- a) More common after life threatening trauma
- b) Seen in seizure and migraine patients

- c) More common in females
- d) Mean age at onset is 25 years

Question 5:

A patient says that he feels that there is something wrong with him mentally and it is because of the summer season. How much would he score on the insight scale?

- a) 2
- b) 3
- c) 4
- d) 5

Question 6:

A patient was prescribed escitalopram. Which of the following adverse effect is the least likely?

- a) Nausea
- b) Vivid dreams
- c) Anorgasmia
- d) Sialorrhea

Question 7:

A 48-year-old male presents to the ER with confusion and shaking of the entire body. He has a history of excessive alcohol consumption. What is the cause of delirium tremens in an alcoholic?

- a) Acute infection
- b) Fatty liver
- c) Gradual withdrawal of alcohol
- d) Small doses of regular consumption of alcohol

Question 8:

A patient of anorexia nervosa was eating all her meals according to the prescribed diet chart but there is no change in her weight after 1 week. Which of the following needs to be done?

- a) Observe the patient for 2 hours post meal
- b) Ask the psychiatrist to prescribe anti-anxiety medication
- c) Increase the calories from 1500 to 2000 in her diet
- d) Increase water intake

Answer Key

Question No.	Correct Option
1	d
2	c
3	a
4	a
5	b
6	d
7	a
8	a

Detailed Explanations

Solution to Question 1:

The provision by which a patient has the right to choose his caretaker and future course of action in treatment because of illness or incapacity is known as an advance directive. It is also known as a living will, personal directive, or medical directive.

Solution to Question 2:

Sleep-onset insomnia is a characteristic feature of anxiety disorder, not melancholia.

Melancholia is one of the oldest terms used in psychiatry, to describe the dark mood of depression. It is still used to refer to a state of depression characterized by:

- Severe anhedonia
- Early morning awakening (terminal insomnia)

- Weight loss
- Profound feelings of guilt (often over trivial events)

Melancholia is sometimes referred to as "endogenous depression" or depression that arises in the absence of external life stressors or precipitants.

Solution to Question 3:

Paradoxical suicide is defined as suicide which occurs when the patient begins to recover from the depression

- This is because, when the patient is in depression, though he/she has the suicidal thoughts cannot act on it due to lack of energy.
- As the patient begins to recover from depression, patients gain energy to put their suicidal thoughts into action.

It is called paradoxical, as one might expect that the risk of suicide reduces as the patient starts to recover, but on contrary to that, the risk of suicide is increased during the recovery phase.

Solution to Question 4:

Depersonalization is common after a life-threatening trauma.

The DSM-V defines depersonalisation as experiences of unreality, detachment, or being an outside observer with respect to one's thoughts, feelings, sensations, body, or actions (e.g., perceptual alterations, distorted sense of time, unreal or absent self, emotional and/ or physical numbing).

Another term called derealization refers to the experiences of unreality or detachment with respect to the surroundings.

The DSM-V diagnostic criteria for depersonalization/derealization disorder is as follows:

- The presence of persistent or recurrent experiences of depersonalization, derealization, or both.
- Intact reality testing, during the episodes.
- The symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- The disturbance is not attributable to the physiological effects of a substance(e.g., a drug of abuse, medication) or another medical condition(e.g., seizures). (option b)
- The disturbance is not better explained by another mental disorder, such as schizophrenia, panic disorder, major depressive disorder, acute stress disorder, posttraumatic stress disorder, or another dissociative disorder.

Characteristic features of depersonalization/derealization disorder:

- The condition commonly occurs after a life-threatening trauma.
- There is no gender preponderance. Male:Female ratio is 1:1 (option c).

- Mean age at onset is 16 years (option d).
- The condition is commonly associated with symptoms such as an altered sense of time, difficulty in remembering past memories, anxiety and depression.

Solution to Question 5:

The patient would score 3, as he is aware of being sick but has attributed it to external factors.

Insight is rated on a 5 point scale:

- Complete denial of illness.
- Slight awareness of being sick and needing help but denying it at the same time.
- Awareness of being sick, but is attributed to physical or external factors.
- Intellectual insight - awareness of being ill, and that the symptoms are due to own particular irrational feelings and thoughts.
- Emotional insight - the highest level of insight. Intellectual insight with making significant changes in future behavior.

Loss of insight is seen in schizophrenia, mania, delusional disorders, etc.

Solution to Question 6:

Sialorrhea is not a side effect of escitalopram.

Escitalopram is a selective serotonin reuptake inhibitor (SSRI). SSRIs are associated with dry mouth, and not sialorrhea.

System	Side effects of SSRIs
Sexual	Anorgasmia, Decreased libido
Gastrointestinal (5-HT ₃ receptor)	Nausea, Diarrhea, Anorexia, Vomiting, Flatulence, and Dyspepsia
Nervous system	Extremely vivid dreams or nightmares, Anxiety, Insomnia/sedation, Seizures, Emotional blunting
Anticholinergic	Dry mouth, Constipation, Sedation
Cardiovascular	Drug induced long QT syndrome
Hematological	Easy bruising
Endocrine	Mammary hyperplasia, Galactorrhea

Solution to Question 7:

Among the given options, acute infection can be a cause of delirium tremens in an alcoholic patient.

Physical illness e.g., hepatitis or pancreatitis predisposes to the syndrome; a person in good physical health rarely has delirium tremens during alcohol withdrawal.

The syndrome usually develops on the third day of admission. A patient admitted for an unrelated condition may unexpectedly have an episode of delirium, which is the first sign of a previously undiagnosed alcohol-related disorder.

Episodes of delirium tremens usually begin at the age of 30-40 years, after 5 to 15 years of heavy drinking, and are typical after a binge. Gradual withdrawal of alcohol does not cause delirium tremens. However, sudden withdrawal can cause it.

Treatment of alcohol withdrawal:

- For patients showing withdrawal symptoms, chlordiazepoxide 25-50 mg every 2-4 hours can be administered orally.
- For patients with delirium tremens, 50-100 mg of chlordiazepoxide should be administered every 4 hours.
- If oral administration of drugs is not possible, intravenous lorazepam can be tried.

Solution to Question 8:

Observe the patient for 2 hours post-meal because the patient might vomit the food.

Treatment of anorexia nervosa:

- Hospitalization:
- Patients with a weight 20% less than expected weight for height should be considered for hospitalization.
- During a hospitalized treatment program, patients should be weighed daily, early in the morning after emptying the bladder.
- The daily fluid intake and urine output should be recorded.
- If vomiting is occurring, hospital staff members must monitor serum electrolyte levels regularly and watch for the development of hypokalemia.
- Because food is often regurgitated after meals, the staff may be able to control vomiting by making the bathroom inaccessible for at least 2 hours after meals or by having an attendant in the bathroom to prevent the opportunity for vomiting.
- Psychotherapy:
- Cognitive-behavioral therapy (CBT)

- Dynamic psychotherapy
- Family therapy
- Pharmacotherapy:
 - Pharmacological agents are not the first-line treatment but cyproheptadine, amitriptyline, clomipramine, pimozide, and chlorpromazine show some improvement in these patients.
 - Trials of fluoxetine have resulted in some reports of weight gain.

Psychiatry AIIMS 2020 (May and Nov INI-CET)

Question 1:

You are attending physiology viva. The examiner asks you who recorded the first EEG. You say "I don't know ". The examiner feels bad for you and gives you a clue that "the name sounds like something you get commonly in McDonalds". You remember the name and answer correctly. What type of memory has been implicated here?"

- a) Semantic memory
- b) Implicit memory
- c) Associative learning
- d) Priming

Question 2:

A known alcoholic patient is brought to the emergency in a disoriented state. He complains of seeing snakes and insects crawling on his skin. Which of the following should be given to this patient?

- a) Haloperidol + thiamine
- b) Lorazepam only
- c) Lorazepam + thiamine
- d) Haloperidol only

Question 3:

A drug that selectively reduces the reuptake of serotonin would likely cause which of the following side effects?

- a) Constipation
- b) Sexual dysfunction
- c) Sialorrhea
- d) Visual disturbance

Question 4:

A patient undergoing ECT for his severe depression was given thiopentone and succinylcholine. The purpose of these medications was to?

- a) To bring calming effect
- b) To produce mood elevation
- c) To induce anesthesia and relax muscles
- d) To prevent memory loss after the ECT

Question 5:

A young male who is shy and prefers social isolation appears to be emotionally cold and resides alone. There is no history of hallucinations or delusions. The most likely diagnosis is

- a) Paranoid personality disorder
- b) Schizoid personality disorder
- c) Antisocial personality disorder
- d) Emotionally unstable personality disorder

Question 6:

A 40-year-old male patient comes to the psychiatry OPD with complaints of repetitive thoughts that, he always feels his hands are dirty, though they are not. This gives him discomfort and hence he has to wash them again and again. This is a disorder of thought

- a) Flow
- b) Form
- c) Possession
- d) Content

Question 7:

Which of the following is not a type of subcortical dementia?

- a) Pick's disease
- b) Parkinson's disease
- c) Huntington's disease

- d) Wilson's disease

Question 8:

Which of the following is not a good prognostic factor in schizophrenia?

- a) Positive symptoms
- b) Late age of onset
- c) Insidious onset of symptoms
- d) Associated with depression

Question 9:

A 7-year-old child presented with complaints of bed-wetting almost twice a week for the past 1 year. A detailed investigation was done to rule out the organic cause. What should be the treatment plan for this child?

- a) Pharmacotherapy with imipramine
- b) Psychodynamic psychotherapy
- c) Bladder training, with reward for delaying micturition during daytime
- d) Bell and pad based classical conditioning

Question 10:

A male patient who lost his job recently became irritable and had a sad mood for most of the time. Thinking about the future worried him and he felt irritation towards his family members. He occasionally went out to watch movies with his friends and was able to enjoy but on returning home he felt similar symptoms. What is the probable diagnosis?

- a) Generalised anxiety disorder
- b) Adjustment disorder
- c) Mixed anxiety with depression
- d) Moderate depression

Question 11:

A 10-year-old child is suffering from selective mutism. He is most probably suffering from?

- a) Childhood depression
- b) Childhood psychosis
- c) Childhood anxiety disorder
- d) Hyperkinetic disorder

Question 12:

A psychotic patient developed voluntary purposeless movements and was once observed to be standing still for hours with waxy flexibility and negativism. What is the appropriate medical management?

- a) Haloperidol
- b) Lorazepam
- c) Clonidine
- d) Propranolol

Question 13:

Which of the following features differentiate delirium from dementia in Alzheimer's disease?

- a) Acuity of onset and agitation
- b) Acuity of onset and loss of consciousness
- c) Visual hallucinations and impaired memory
- d) Agitation and irritation

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	c
5	b
6	c
7	a

8	c
9	d
10	b
11	c
12	b
13	b

Detailed Explanations

Solution to Question 1:

In the given scenario, priming is the most appropriate answer.

The examiner is asking for a fact, and then prompting the answer, by giving a clue which sounds (phonological) like the answer. This is an example of priming.

Prime is any cue, which precedes the recall of the target word. It maybe similar to hear (sun-bun-gun) or may hold similarity to the target word (Sun- moon - star).

Solution to Question 2:

The given clinical scenario points towards delirium tremens likely due to acute alcohol withdrawal. Lorazepam + thiamine should be given to this patient.

Delirium tremens occurs after 2-4 days of abstinence in a heavy or chronic drinker. It is characterized by autonomic hyperactivity, tremors, delusions, and hallucinations. It usually resolves spontaneously in 7 days but may cause the death of the patient due to cardiovascular collapse or infections or self-injury.

Delirium tremens is treated using benzodiazepines starting with a higher dosage which is gradually tapered as withdrawal symptoms disappear. In cases of chronic alcohol use, vitamin deficiency (especially thiamine) may be co-existing and hence bolus doses of thiamine are given to prevent/treat Wernicke encephalopathy.

Haloperidol is generally not the first-line drug in case of alcoholic withdrawal. It might be used as an add-on in agitated patients. It decreases the seizure threshold and its use would be appropriate in a case of co-existing psychotic disorder like schizophrenia.

Drug Therapy For Alcohol In toxication and Withdrawal			
Clinical Problem	Drug	Route	Dosage
Tremulousness and mild to moderate agitation	Chlordiazepoxide Diazepam	Oral	25-100 mg every 4-6 hrs 5-20 mg every 4-6 hrs
Hallucinosiis	Lorazepam	Oral	2-10 mg every 4-6 hrs

Drug Therapy For Alcohol Intoxication and Withdrawal			
Extreme agitation	Chlordiazepoxide	Intravenous	0.5 mg/kg at 12 mg/min
Withdrawal seizures	Diazepam	Intravenous	0.15 mg/kg at 2.5 mg/min
Delirium tremens	Lorazepam	Intravenous	0.1 mg/kg at 2.0 mg/min

Solution to Question 3:

Selective serotonin reuptake inhibitors (SSRIs) are associated with sexual dysfunction.

Note - Sexual dysfunction is the most prominent side effect of SSRIs and weak anticholinergic effects are only shown by paroxetine, and the GI effects of SSRIs predominantly include diarrhoea.

System	Side effects of SSRIs
Sexual	Anorgasmia, Decreased libido
Gastrointestinal (5-HT ₃ receptor)	Nausea,diarrhea, anorexia, vomiting, flatulence, and dyspepsia
Nervous system	Extremelyvivid dreamsor nightmares, Anxiety, Insomnia/sedation, seizures, emotional blunting
Anticholinergic	Dry mouth, constipation, sedation
Cardiovascular	Drug-induced long QT syndrome
Hematological	Easy bruising
Endocrine	Mammoplasia, galactorrhea

Solution to Question 4:

Thiopentone and succinylcholine are given for anesthesia induction and muscle relaxation respectively.

Thiopentone sodium is a barbiturate and acts by opening chloride channels, whilst succinylcholine is a depolarising muscle relaxant.

ECT can be of 2 types:

- Direct - without anesthesia and muscle relaxant
- Modified - with anesthesia and muscle relaxant

Currently modified ECT is preferred and is given with adequate spacing i.e., no more than 3 ECT sessions per week.

After adequate preparation, a current of 200-1600mA (110V for 0.1-1 seconds) is given, resulting in an adequate seizure. Electroconvulsive therapy treatments are associated with retrograde amnesia for a period of several minutes before the treatment and anterograde amnesia after the treatment.

Solution to Question 5:

The diagnosis, in this case, is schizoid personality disorder.

Schizoid personality disorder is a cluster A personality disorder. The patient neither desires nor enjoys close relationships (shy and prefers social isolation, resides alone as in the question) and shows emotional coldness and detachment.

In contrast, subjects with emotionally unstable (borderline) personality disorder cannot tolerate being alone, and they search for companionship.

Solution to Question 6:

The clinical history is suggestive of obsessive-compulsive disorder, which is a disorder of thought possession.

The diseases under thought possession include:

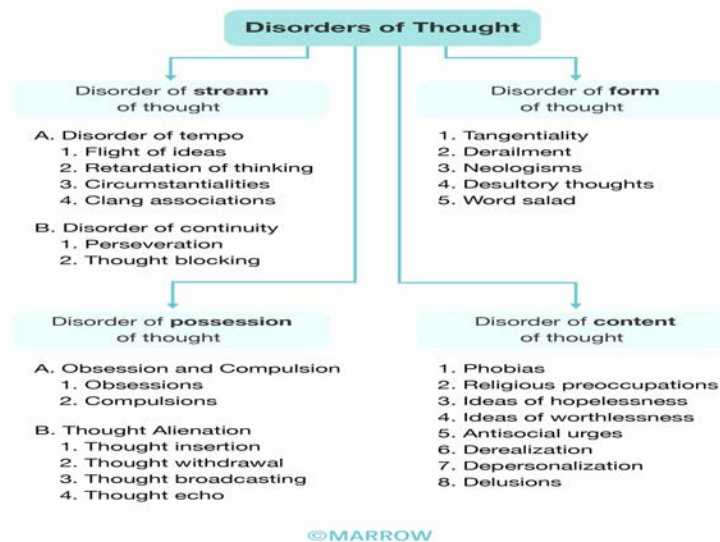
- Obsession and Compulsion
- Thought Alienation

In obsessions and compulsions, although the sense of thought possession in the patient is not impaired, he/she experiences the obsessive thoughts to appear against their will. Hence, despite the patient perceiving a loss of control over the voluntary act (compulsion), they still retain personal possession (ownership) of the act.

Obsessive-compulsive disorder is defined as the presence of obsessions and/or compulsions.

- Obsessions are recurrent and persistent ideas, thoughts, urges, or images that are experienced as intrusive and unwanted and cause anxiety or distress to the patients.
- Compulsions are the repetitive and intentional behaviors or acts performed in response to obsessions or according to certain rules that must be applied rigidly (e.g., repetitive hand washing, ritualistic checking).

In the given case, the obsessional thoughts are the repetitive, unwanted thoughts that his hands are dirty, while the compulsion is to wash his hands to neutralize the obsession.



Solution to Question 7:

Pick's disease (a type of frontotemporal dementia) is a type of cortical dementia.

	Cortical dementia	Subcortical dementia
Site of lesion	Cortex	Thalamus, basal ganglia, cerebellum, brain stem
Higher cortical symptoms	AphasiaApraxiaAgnosiaProsopagnosiaAcalculia	No
Motor symptoms	No	DysarthriaTremorsAtaxiaRigidity
Memory	Recall and recognition are impaired	Recall is impaired
Examples	Alzheimer's diseaseFrontotemporal dementiaCreutzfeldt Jakob disease	Parkinson's diseaseProgressive supranuclear palsyHuntington's diseaseWilson's diseaseIdiopathic basal ganglia calcificationNormal-pressure hydrocephalusVascular dementia

Solution to Question 8:

Insidious onset of symptoms is a poor prognostic factor of schizophrenia.

Prognostic factors of Schizophrenia:

T.J.Crow classified schizophrenia into two separate types :

Good Prognostic factors	Poor prognostic factors
Acute onset	Insidious onset
Late-onset	Early-onset
Presence of a stressor	Absence of a stressor
Short duration < 6 months	Duration >2 years
Presence of depression	No depression
Positive symptoms	Negative symptoms
Family history of mood disorder	Family history of schizophrenia
Married, female sex	Single, male sex
Catatonic schizophrenia	Hebephrenic schizophrenia
Type 1 schizophrenia	Type 2 schizophrenia
	No remissions in 3 years
	History of perinatal trauma
	History of assaultiveness

Type 1	Type 2
Positive symptoms present	Negative symptoms present
More responsive to treatment	Less responsive to treatment
No demonstrable neuropathology	Demonstrable neuropathology (CT/MRI)
Good prognosis	Poor prognosis

Solution to Question 9:

The clinical features are suggestive of nocturnal enuresis. Bell and pad based classical conditioning is the most effective treatment.

Nocturnal enuresis refers to the occurrence of involuntary urination at night after 5 years of age. It may be a normal phenomenon in children up to 5 years of age.

Treatment for nocturnal enuresis may include:

Behavioural therapy:

- Classic conditioning with the bell (or buzzer) and pad (alarm) apparatus is the most effective treatment for enuresis.
- Dryness is achieved in 50% of the cases.
- Bladder training with encouragement or reward for delaying micturition for increasing times during waking hours is useful but considered inferior to bell and pad.

Pharmacotherapy:

- Medications should be considered if behavioural therapy has failed, and if enuresis is causing impairment in social, family, and school function.
- Desmopressin (an antidiuretic) and Reboxetine (norepinephrine reuptake inhibitor) are commonly used drugs. Reboxetine is considered as a safer alternative to imipramine.

Psychotherapy:

- It is indicated in cases with chronic enuresis, to deal with any coexisting psychiatric problems and emotional and family difficulties.

Solution to Question 10:

The clinical features are suggestive of an adjustment disorder.

Adjustment disorder is an emotional response to a stressful event. Stressors may be single like loss of job or divorce; multiple like the death of a parent together with loss of a job; or recurrent like seasonal business difficulties.

Diagnosing criteria for adjustment disorder:

- Development of emotional or behavioural response within 3 months of the onset of the stressor.
- The symptoms or behaviour are clinically significant, where, any one of the following are met:
- The response is out of proportion to the severity or intensity of the stressor.
- Significant impairment in social, occupational, or other areas of functioning.
- The stress-related disturbance does not meet the criteria for another mental disorder or exacerbation of pre-existing mental disorder.
- Mostly depressive but can also have anxiety symptoms.
- The symptoms disappear within 6 months if the stressor is removed.

In the given case, it has been mentioned that the patient lost his job recently. Although a time period is not mentioned in the question, the term recently can be understood as well within the 3 months duration. The patient's behaviour such as sad and irritated affected his social functioning. As he was finding joy in watching movies, depression can be ruled out.

Solution to Question 11:

The child is most likely suffering from a childhood anxiety disorder.

Selective mutism refers to the lack of speaking (being mute) in one or more specific social situations, commonly in school, but speaking well in other situations, commonly at home. The child may be either completely silent at school or maybe just whispering. Rarely, a child may be silent at home but communicating normally at school.

The disorder may develop in children between 4 to 8 years of age and is associated with a social anxiety disorder. They may have symptoms of separation anxiety disorder, school refusal, and

delayed language acquisition.

The first-line treatment includes cognitive behavioural therapy.

Option A and B: selective mutism is not commonly associated with childhood depression or psychosis

Option D: Hyperkinetic disorder is associated with overfamiliarity.

Solution to Question 12:

The clinical features are suggestive of a catatonic disorder. Benzodiazepines such as lorazepam are effective in the treatment of catatonia.

Catatonia is a syndrome characterized by striking behavioral abnormalities such as motoric immobility or excitement, negativism, or echolalia (mimicry of speech), or echopraxia (mimicry of movement).

The clinical picture may include three or more of the following features:

- Stupor: Absence of psychomotor activity and not actively relating to the environment.
- Catalepsy: Passive induction of a posture held against gravity.
- Waxy flexibility: Slight, even resistance to positioning by examiner.
- Negativism: Opposition or no response to instructions or external stimuli.
- Posturing: Spontaneous and active maintenance of a posture against gravity.
- Mannerism: Odd, circumstantial caricature of normal actions.
- Stereotypy: Repetitive, abnormally frequent, non-goal-directed movements
- Automatic obedience: Commands are followed automatically, irrespective of their nature.
- Grimacing
- Ambitendency: Due to ambivalence, opposing actions are carried out.
- Echopraxia: Mimicking another's movement.
- Echolalia: Mimicking another's speech.
- Mutism: Very little or no verbal response to instructions or external stimuli.
- Verbigeration: Incomprehensible speech.

Catatonic disorder may occur due to:

- Advanced mood or psychotic illnesses
- Neurological disorders (e.g., nonconvulsive status epilepticus, and head trauma)
- Infections (e.g., encephalitis)
- Metabolic disturbances (e.g., hepatic encephalopathy, hyponatremia, and hypercalcemia)
- Drugs induced- like corticosteroids, immunosuppressants, and antipsychotics (i.e., neuroleptic) agents.

Management: The primary treatment includes:

- Identifying and treating underlying medical or pharmacological causes. Any offending substances should be removed.
- Benzodiazepines, especially, lorazepam, are effective in improving the symptoms.
- Electroconvulsive therapy is the definitive treatment in cases of life-threatening catatonia (if the patient is unable to eat) or if the patient has developed malignant catatonia.

Solution to Question 13:

Delirium is characterized by acute onset of symptoms and is often accompanied by diminished or loss of consciousness and cognition. This feature is useful to clinically differentiate between delirium and dementia.

Dementia is a group of diseases marked by chronic, progressive cognitive impairment with clear consciousness.

Psychiatry NEET 2018

Question 1:

A 40-year-old lady met with a car accident 2 months ago, following which she complains of screaming and waking up in the night after having nightmares of the same incident over and over again. Which of the following conditions is she suffering from?

- a) Acute stress reaction
- b) Adjustment disorder
- c) Mania
- d) Post traumatic stress disorder

Question 2:

Which of the following features is more in favour of delirium?

- a) Occurs gradually over a period of time
- b) Fluctuating course
- c) Preserved consciousness
- d) Commonly associated with auditory hallucinations

Question 3:

What is defined as the emotional release and discharge after consciously reliving a painful experience that has been repressed?

- a) Catharsis
- b) Abreaction
- c) Venting out
- d) Guided relaxation

Question 4:

What is 'mental retardation' now called according to the American Psychiatric Association in its revision of DSM-5?

- a) Feeble mindedness
- b) Mental subnormality
- c) Intellectual disability
- d) Intellectually challenged

Question 5:

The confusion assessment test is used in:

- a) Dementia
- b) Depression
- c) Delirium
- d) Schizophrenia

Question 6:

Which of these disorders was earlier known as minimal brain dysfunction?

- a) Dyslexia
- b) Attention deficit/hyperactivity disorder
- c) Mental subnormality
- d) Oligophrenia

Question 7:

Seman's squeeze technique is used for which of the following conditions?

- a) Retrograde ejaculation
- b) Erectile dysfunction
- c) Premature ejaculation
- d) Painful ejaculation

Question 8:

Addison's disease can present with symptoms similar to which of the following disorders?

- a) Depression

- b) Generalised anxiety disorder
- c) Bipolar disorder
- d) Panic disorder

Question 9:

Which of the following is not an OCD related disorder?

- a) Hair pulling disorder
- b) Temper tantrums
- c) Hoarding disorder
- d) Skin picking

Question 10:

Which of the following is not true about Alzheimer's disease?

- a) Intracellular neurofibrillary tangles are seen
- b) Neuritic plaques made of beta amyloid are found
- c) Nucleus of Meynert is not affected
- d) Short term memory is affected

Answer Key

Question No.	Correct Option
1	d
2	b
3	b
4	c
5	c
6	b
7	c
8	a
9	b
10	c

Detailed Explanations

Solution to Question 1:

The given clinical scenario is suggestive of post-traumatic stress disorder. PTSD is a psychiatric disorder occurring after a life-threatening or potentially fatal event, (e.g., war, house fire, serious accident, rape, robbery) affecting the patient or the patient's close friend or relative.

Symptoms can be divided into four types:

- Re-experiencing (e.g., intrusive memories of the event [flashbacks] and nightmares)
- Hyperarousal (e.g., anxiety, increased startle response, impaired sleep, hypervigilance)
- Negative cognition (e.g., negative beliefs, emotional numbing)
- Avoidance (e.g., avoiding stimuli associated with the traumatic event)

Psychotherapies found to be effective for PTSD include exposure therapy, trauma-focused cognitive-behavioral therapy, eye movement desensitization, and reprocessing.

Solution to Question 2:

Fluctuating course is suggestive of delirium.

Characteristic presentation of delirium:

- Acute onset
- Brief duration - usually days to weeks
- Decline in the level of consciousness.
- Fluctuating course - unpredictable fluctuations in severity and other clinical manifestations
- Sundowning - worsening of symptoms at night
- Rapid improvement - when the underlying cause is treated
- Other characteristics include asterixis, carphologia (picking movements on cover sheets and clothes)
- More commonly associated with visual hallucinations

Solution to Question 3:

Abreaction is a psychoanalytical term for reliving an experience to release it emotionally. It is a specific type of catharsis done by becoming conscious of repressed traumatic events.

Catharsis is a general term for the process of releasing strong or repressed emotions. It can be done through various mediums, like expressing emotions through art. When the unconscious repressed memories are consciously recalled and relived, this technique specifically is called abreaction.

Solution to Question 4:

Mental retardation is now known as Intellectual Disability, according to the revised DSM-5, by the American Psychiatric Association.

Solution to Question 5:

The confusion assessment test is used in the detection of delirium.

Delirium is reported to occur in a wide range of mechanically ventilated ICU patients and can be detected by the Confusion Assessment Method (CAM)-ICU or the Intensive Care Delirium Screening Checklist.

These tools are used to ask patients to answer simple questions and perform simple tasks and can be used readily at the bedside.

Solution to Question 6:

Attention deficit/hyperactivity disorder (ADHD) was known as minimal brain dysfunction earlier.

Solution to Question 7:

Seman's squeeze technique is used for premature ejaculation, to prolong the time period until ejaculation.

Seman's technique (stop-start technique): It is a variation of the squeeze technique where a man approaches the point close to ejaculation and then stops before ejaculation and relaxes before starting stimulation again. This method prolongs the time until ejaculation by relying on behavior modification.

Other treatments for premature ejaculation:

- Squeeze technique (Master and Johnson's technique)- as a man approaches a climax, he squeezes the tip of the penis just below the head of the penis. The pressure is held there until the sensation of impending orgasm diminishes. The process can then be started over again so that over time a man prolongs the time period until he reaches ejaculation.
- Sensate focusing- refocussing on the non-coital aspects of sexual activity.
- Selective serotonin reuptake inhibitors (SSRIs)- delays ejaculation. e.g.- dapoxetine, paroxetine.

Solution to Question 8:

Addison's disease presents most similarly to depression. Apathy, social withdrawal, impaired sleep, decreased concentration, and prominent fatigue is frequently seen which is usually mistaken for depression.

Following infection, trauma, or surgery, a patient with previously unrecognized Addison's disease may suffer from an Addisonian crisis. Such patients present with either shock or hypoglycemia. This is a medical emergency requiring urgent medical referral.

Endocrinopathy	Differential psychiatric diagnosis
Addison's disease	Depression
Hyperthyroidism	Mood disorder, panic disorder, generalized anxiety disorder
Hypothyroidism	Depression, hypomania, delusions, schizophrenia (myxedema madness).
Hypoparathyroidism	Delirium is the most common.
Hyperparathyroidism	Depression, anxiety, paranoid delusions. Tetrad of "painful bones, renal stones, abdominal groans, and psychiatric overtones".
Hypoglycemia	Anxiety, depersonalization, derealisation, aggressive behavior, amnesia.

Solution to Question 9:

Temper tantrums are not OCD-related.

Obsessive-compulsive and related disorders are:

- Obsessive-Compulsive Disorder
- Body Dysmorphic Disorder
- Hoarding Disorder
- Hair-Pulling Disorder (Trichotillomania)
- Excoriation (Skin-Picking) Disorder

Solution to Question 10:

Nucleus of Meynert is affected in Alzheimer's disease.

Biochemically, there is a reduction of acetylcholine and degeneration of cholinergic neurons in the nucleus basalis of Meynert that project throughout the cortex. There is also noradrenergic and serotonergic depletion due to degeneration of brain stem nuclei such as the locus ceruleus and the dorsal raphe.

Microscopically, there are neuritic plaques containing amyloid- β , neurofibrillary tangles, and amyloid- β accumulation in the blood vessel walls of the cortex and leptomeninges.

The FDA recently approved Flortaucipir F18 for intravenous injection, the first radioactive diagnostic agent for adult patients with cognitive impairment who are being evaluated for Alzheimer's disease. It is indicated for PET imaging of the brain to estimate the density and distribution of aggregated tau neurofibrillary tangles (NFTs), a primary marker of Alzheimer's disease.

Psychiatry NEET 2019

Question 1:

Which of the following drugs is used as an anti-smoking agent?

- a) Busulfan
- b) Acamprosate
- c) Varenicline
- d) Gabapentin

Question 2:

What is the false statement regarding somnambulism?

- a) Disorder of sleep arousal
- b) Person will have full consciousness
- c) Usually, terminates in awakening followed by confusion
- d) There may be neurological condition associated

Question 3:

Which of the following is not a risk factor for delusional disorder?

- a) Recent immigration
- b) Social isolation
- c) Family history
- d) Young age

Question 4:

Premature ejaculation is a disorder of which phase of the normal sexual cycle?

- a) Desire
- b) Arousal
- c) Orgasm

d) Pain

Question 5:

Confusion assessment test is used in:

- a) Delirium**
- b) Dementia**
- c) Schizophrenia**
- d) Depression**

Question 6:

Which of the following has a poor prognosis with exposure and response prevention in OCD?

- a) Pathological doubt**
- b) Magical thinking**
- c) Hoarding**
- d) Contamination obsession**

Question 7:

What is the fear of heights known as?

- a) Acarophobia**
- b) Algophobia**
- c) Agoraphobia**
- d) Acrophobia**

Answer Key

Question No.	Correct Option
1	c
2	b
3	d

4	c
5	a
6	c
7	d

Detailed Explanations

Solution to Question 1:

Varenicline is an $\alpha 4\beta 2$ nicotinic acetylcholine receptor partial agonist used in the treatment of nicotine addiction.

The FDA approved medication to aid in smoking cessation include:

- NRT-Nicotine Replacement Therapy
- Bupropion
- Varenicline

Drugs are only adjunctive to behavioral therapy, which is the first line of management.

Behavior therapy is the most widely accepted and well-proven psychological therapy for treating Tobacco use disorder. It consists of several techniques:

- Skill training and relapse prevention.
- Stimulus control.
- Mindfulness approaches.

Bupropion:

- It enhances both noradrenergic and dopaminergic neurotransmission by inhibiting reuptake.
- Used for aiding smoking cessation and for the treatment of depression in combination with SSRIs.
- Side effects: seizures, sedation

Varenicline:

- $\alpha 4\beta 2$ Nicotinic ACh receptor partial agonist.
- Used for smoking cessation.
- Side effects: sleep disturbance, may depress mood and increase suicidal tendencies.

Mirtazapine:

- Serotonin receptor antagonist.
- Used for the treatment of depression in combination with SSRIs and for insomnia in depressive patients.
- Side effect: sedation

Solution to Question 2:

Somnambulism is characterized by sequence of behaviors without full consciousness.

Note: About 80% of sleepwalkers have a family history of sleepwalking, and an identical twin is much more likely to sleepwalk if their other twin sleepwalks.

Solution to Question 3:

Advanced age is a risk factor for delusional disorders, not young age.

Risk factors for delusional disorders:

- Advanced age
- Sensory impairment or isolation
- Personality features (e.g.unusual interpersonal sensitivity)
- Family history
- Social isolation
- Recent immigration

Solution to Question 4:

Premature ejaculation is a disorder of the orgasm phase of the normal sexual cycle.

Premature ejaculation is when ejaculation occurs before the completion of satisfactory sexual activity. In severe cases, it is characterized by ejaculation either before penile entry into the vagina or soon after penetration. The causes can be biological or psychological, like performance anxiety.

Solution to Question 5:

Confusion assessment test is used in detection of delirium.

Delirium is reported to occur in a wide range of mechanically ventilated ICU patients and can be detected by the Confusion Assessment Method (CAM)-ICU or the Intensive Care Delirium Screening Checklist. These tools are used to ask patients to answer simple questions and perform simple tasks and can be used readily at the bedside.

Solution to Question 6:

Hoarding behavior shows poor prognosis with exposure and response prevention.

Challenges posed by hoarding patients to typical cognitive behavior therapy (CBT) include poor insight into the behavior, low motivation, and resistance to treatment.

The most effective treatment for hoarding is a cognitive-behavioral model that includes:

- Training in decision making and categorizing
- Exposure and habituation to discarding
- Cognitive restructuring.

Solution to Question 7:

Acrophobia is the extreme or irrational fear of heights.

Phobia	Fear of
Acarophobia	Small bugs
Acrophobia	High places
Agoraphobia	Open or crowded places, leaving familiar setting of home
Aichmophobia	Pointed objects
Ailurophobia	Cats
Algophobia	Pain
Androphobia	Men
Astraphobia or astrapophobi a	Thunder and lightning
Autophobia	Being alone, or of self
Bathophobia	Depths
Claustrophobia	Closed or confined places
Coprophobia	Excreta
Cynophobia	Dogs
Demophobia	Crowds
Erythrophobia	Blushing
Gynophobia	Women
Haphephobia	Being touched
Hematophobia	Sight of blood
Hydrophobia	Water
Hypnophobia	Sleep
Lalophobia or glossophobia	Speaking
Mysophobia	Dirt, contamination

Phobia	Fear of
Necrophobia	Dead bodies
Nyctophobia	Darkness, night
Panphobia	Everything
Pathophobia or nosophobia	Disease, suffering
Peccatophobia	Sinning
Pedophobia	Children
Phonophobia	Speaking aloud
Photophobia	Strong light
Sitophobia	Eating
Taphophobia	Being buried alive
Thanatophobia	Death
Toxophobia	Being poisoned
Xenophobia	Strangers
Zoophobia	Animals

Psychiatry NEET 2020

Question 1:

A 25-year-old male presented to the clinic with a complaint of the feeling of insects crawling under the skin. Which of the following drug abuse can cause the symptoms of this patient?

- a) Cannabis
- b) Cocaine
- c) Amphetamine
- d) Alcohol

Question 2:

A 60-year-old male who lost his wife 3 months back complains that his intestines have become rotten. He feels that he is responsible for his wife's death and should be sent to prison. He complains of feeling low all the time and has lost interest in daily activities since his wife's death. What is the most likely diagnosis?

- a) Normal grief reaction
- b) Psychotic depression
- c) Delusional disorder
- d) Schizophrenia

Question 3:

What is the treatment of choice for postpartum blues?

- a) Fluoxetine
- b) Cognitive behavioural therapy
- c) Lithium carbonate
- d) Support and education

Question 4:

A 25-year-old woman complains of a creepy crawling sensation in her legs. It is more at night and prevents her from sleeping. She is relieved of these symptoms by either walking or moving her legs. Which of the following drugs is used in treating the condition?

- a) Gabapentin
- b) Pramipexole
- c) Vitamin B12
- d) Iron tablets

Question 5:

A first-year MBBS student came to OPD with intermittent episodes of a feeling of impending doom accompanied by intense perspiration. Usually, the episodes occur prior to her exams. What is the most likely diagnosis?

- a) Panic disorder
- b) Conversion disorder
- c) Social anxiety disorder
- d) Generalized anxiety disorder

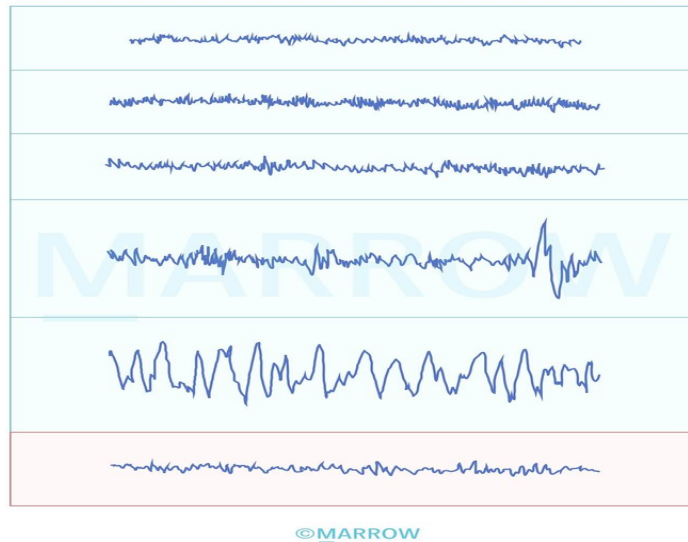
Question 6:

What is the maximum period of voluntary self-admission under the Mental Healthcare Act 2017?

- a) 48 hours
- b) 7 days
- c) 30 days
- d) 90 days

Question 7:

Identify the stage of sleep based on EEG finding highlighted in red.



- a) Stage 1
- b) Stage 2
- c) Stage 3
- d) REM sleep

Question 8:

A middle-aged female patient presents with a history of loss of interest in day to day activities, weight loss, insomnia, and sadness for six months. She also recalls hearing a voice once that ordered her to kill herself. What is the DSM-V diagnosis?

- a) Major depressive disorder with psychotic features
- b) Schizophrenia
- c) Schizoaffective disorder
- d) Delusional disorder

Answer Key

Question No.	Correct Option
1	b
2	b
3	d

4	a
5	a
6	c
7	d
8	a

Detailed Explanations

Solution to Question 1:

The feeling of insects crawling under the skin, also known as cocaine bugs or formication, is a feature of cocaine abuse.

Other features of acute cocaine intoxication are delusions of persecution, paranoia, and suspiciousness. It is essential to differentiate acute cocaine intoxication from acute schizophrenia which it closely resembles.

Solution to Question 2:

The given clinical scenario of persistent low mood, anhedonia, guilt, and self-critical thought content along with nihilistic delusion (rotten intestine), points towards a diagnosis of major depression with psychotic features.

Differentiating features of grief and depression have been enlisted in the following table:

Note: Tricyclic antidepressants are considered to be the first-line drugs for psychotic depression.

Grief	Depression
Feelings of emptiness and loss	Persistent low mood and inability to anticipate pleasure
Dysphoria comes in waves and tends to decrease in intensity over time	Low mood tends to be persistent and consistent
Associated with thoughts and reminders of deceased/loss	Not tied to specific thoughts and pre-occupations
May be present along with positive emotions and humor	Pervasive unhappiness and misery
Thought content associated with thoughts and memories of deceased/loss	Thought content associated with self-critical and pessimistic ruminations
Self-esteem is generally preserved	Worthlessness and self-loathing is common

Grief	Depression
Ideas of death focused on joining the deceased	Ideas of death focused on ending life due to hopelessness

Solution to Question 3:

Postpartum blues is common in mothers in the puerperal period and requires nothing more than education and support. It is characterised by a labile mood, sadness and subjective confusion. It is thought to be due to changes in hormone levels, the stress of childbirth and anxiety about the responsibilities of motherhood.

If symptoms last longer than 2 weeks, an evaluation for postpartum depression is needed.

Solution to Question 4:

The history is suggestive of restless leg syndrome and gabapentin is the first-line choice of treatment for this condition.

Gabapentinoids are considered superior to dopamine agonists due to the high frequency of intermediate and long-term augmentation with dopamine agonists, as well as their risk of impulse control disorders.

Restless legs syndrome (RLS) is an irresistible urge to move the legs when at rest or while trying to fall asleep or an uncomfortable sensation of creeping-crawling feeling. It is worse at night and moving legs or walking reduces discomfort. Uremia, neuropathies, as well as iron-deficiency and folic acid-deficiency anemias can produce secondary RLS.

In patients with restless legs syndrome, iron stores are initially repleted, and the aggravating factors are eliminated. If RLS still persists, initial therapy with a gabapentinoid (pregabalin, gabapentin, gabapentin enacarbil) is preferred, rather than a dopamine agonist (pramipexole, ropinirole, rotigotine).

FDA-approved drugs for the treatment of restless leg syndrome:

- Gabapentin
- Ropinirole
- Pramipexole
- Rotigotine

Note: A dopamine agonist is used as an alternative in patients with increased risk for side effects from gabapentinoids, such as those with obesity and its complications, a past or present history of moderate to severe depression, a disorder causing gait instability or respiratory failure, and a history of substance use disorder.

Solution to Question 5:

The most likely diagnosis for this patient is panic disorder.

A panic attack is an abrupt surge of intense fear or intense discomfort. Recurrent panic attacks for more than one month is classified as panic disorder.

Panic attacks can occur in mental disorders other than panic disorder, particularly in specific phobia, social phobia, and PTSD.

Solution to Question 6:

The maximum period of voluntary self-admission is 30 days.

When a person has a mental illness that requires admission, and if he/she is able to make his or her own decision and voluntarily requested admission, then it is called voluntary self-admission.

According to the Mental Healthcare Act, the maximum duration for which such admission can last is 30 days.

If the admission requires more than 30 days, or if it requires readmission within 7 days after discharge, then such patients should be examined by two psychiatrists and can be admitted for a maximum period of 90 days.

Note: The aforementioned question is as per the 2020 recall. This concept has been updated, and the admissions are classified as independent and supported. Please refer to the MCQ MG6497 for recent information.

Solution to Question 7:

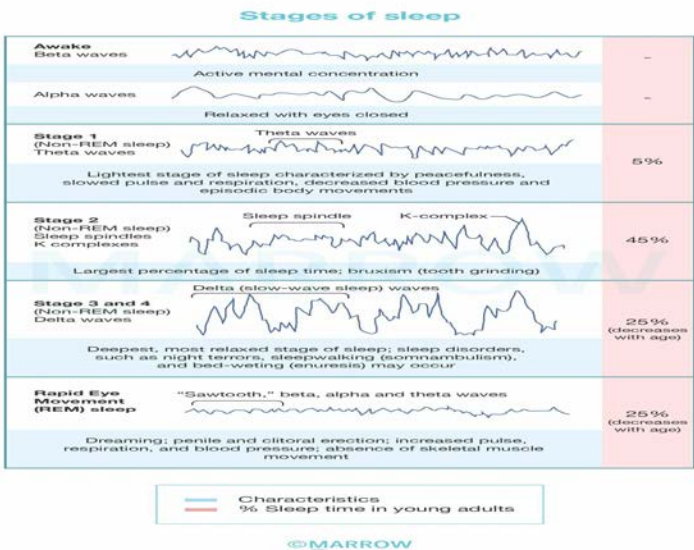
The marked area in the given EEG shows sawtooth waves that are characteristic of REM sleep.

Sleep is comprised of two distinct physiological states, non-rapid eye movement (NREM) sleep and REM sleep. Stages I to IV make up the NREM sleep with marked lower physiological functions than during wakefulness. Dreams which occur during this state are usually lucid and purposeful.

REM sleep is characterized by a high level of brain activity. The physiological activity is similar to that during wakefulness. In REM sleep,

- Pulse, respiration, and blood pressure are increased.
- Brain oxygen use increases during REM sleep.
- Thermoregulation is altered - Poikilothermia prevails during REM sleep.
- There is loss of muscle tone - body movements are absent during REM sleep.
- There is partial or full penile erection - This is used to evaluate the cause for impotence.
- Dreams are abstract and surreal.

In normal adults, a REM episode has a latency of about 90 minutes during which the sleep is in the NREM stage. This latency is shortened in narcolepsy and depressive disorders.



Solution to Question 8:

Since the patient fulfills the DSM-V criteria for major depressive disorder and has had an episode of auditory hallucination, a diagnosis of major depressive disorder with psychotic features is made.

Diagnostic criteria for depression:

- Depressed mood most of the day
- Anhedonia- diminished interest in day to day activities
- Significant weight loss or weight gain. Change in appetite.
- Insomnia or hypersomnia
- Psychomotor agitation or retardation
- Fatigue or loss of energy
- Feelings of worthlessness or excessive, inappropriate guilt
- Diminished ability to think or concentrate, indecisiveness
- Recurrent thoughts of death, suicidal ideation

At least five out of nine symptoms must be present over a two week period and one symptom must be depressed mood or anhedonia.

In major depressive disorder with psychotic features, delusions and/or hallucinations are present along with the depression symptoms.

Psychiatry INI-CET 2021

Question 1:

A person often feels that his hands are contaminated and he is forced to wash his hands. Recently, he feels this repetitive, distressing thought of repetitive hand washing has begun affecting his performance. Which of the following is the best treatment option for this patient?

- a) DBT
- b) Exposure and response prevention
- c) Systematic desensitization
- d) CBT

Question 2:

Match the following

- a) A-3,B-4,C-1,D-2
- b) A-4,B-3,C-1,D-2
- c) A-3,B-4,C-2,D-1
- d) A-2,B-1,C-3,D-4

Question 3:

Which of the following is true regarding nicotine substitution therapy?

- a) Preferably given by gastrointestinal route
- b) Varenicline comes with a blackbox warning of cardiovascular death
- c) There should be a 15 minute gap between nicotine gum and coffee/soda/acidic food as they decreases its absorption
- d) Nicotine chewing gum is better for constant use as it gives 25% higher blood level than lozenges

Question 4:

Match the following

- a) A-4, B-2, C-3, D-1
- b) A-3, B-4, C-1, D-2
- c) A-1, B-4, C-2, D-3
- d) A-2, B-4, C-1, D-3

Question 5:

Which of the following is not classified as OCD as per ICD 11?

- a) Trichotillomania
- b) Body dysmorphic disorder
- c) Hypochondriac disorder
- d) PTSD

Question 6:

Which of the following is a true statement about obsessive compulsive disorder (OCD)?

- a) Atypical antipsychotics is the first line treatment
- b) Prevalence is 7 -10 percent in general population
- c) Depression is a common comorbidity
- d) Contamination is uncommon obsession

Question 7:

Which of the following is true statement about Korsakoff syndrome?

- a) Triad of confusion, ophthalmoplegia, and ataxia
- b) Loss of long term memory but intact personality
- c) Anterograde amnesia with loss of recent memory
- d) Loss of implicit memory and immediate recall

Answer Key

Question No.	Correct Option
1	b
2	c
3	c
4	d
5	d
6	c
7	c

Detailed Explanations

Solution to Question 1:

A patient presenting with an obsession with contamination and repetitive handwashing is suggestive of classical obsessive-compulsive disorder (OCD) and the best treatment option for this patient is exposure and response prevention (ERP).

First line management of OCD in adults can include pharmacotherapy or psychotherapy. Psychotherapy is primarily based on exposure and response prevention.

Exposure and response prevention is a form of exposure therapy that is included under behavioral therapy. It is a more specific treatment modality for OCD than CBT. It refers to exposing the patient to OCD-inducing situations and preventing them from exhibiting the usual compulsive responses. The level of exposure and control of response to it, is gradual and is decided by the therapist and the patient. The dose and duration of exposure and response prevention therapy should be sufficient for it to be effective.

Cognitive therapies aim to address distortions in thought like increased responsibility to clean hands by replacing them with adaptive thinking patterns.

Pharmacotherapy consists of SSRIs and clomipramine. Pharmacotherapy is preferred in patients with severe symptoms or with co-morbid depression.

Combined therapy with Pharmacotherapy+ Psychotherapy are effective in patients not responding to either of them alone.

Somatic therapies like targeted neurosurgical lesions to cortico-striatal-thalamic cortical tracts (CSPC) and deep brain stimulation have proven efficacious in treatment-refractory patients.

Cognitive behavioral therapy (Option D) with ERP for OCD is preferred in children and adolescents. Cognitive-Behavioral therapy comprises of a number of treatment patterns that include both cognitive and behavioral interventions. Cognitive therapy aims to correct maladaptive beliefs and behavioral therapy facilitates better functioning by reducing symptoms. Broadly, it is used to treat depression, generalized anxiety disorder, post traumatic stress disorder, panic disorder, eating disorders, substance use disorder, obsessive compulsive disorder, insomnia, smoking, low back pain.

Other options:

Option A: Dialectical behavioral therapy (DBT) is a long-term behavioral intervention that is used in borderline personality disorder and to reduce suicidal behavior in adults.

Option C: Systemic desensitization is counterconditioning the patient by approaching the feared situation gradually, in a relaxed state. It is useful in phobias, obsessions, compulsions, and some sexual disorders. It includes relaxation training, hierarchy construction and desensitization of the stimulus.

Solution to Question 2:

The correct option is A-3,B-4,C-2,D-1.

- Depression - Difficult to diagnose due to the shell of physician's pride
- Suicide - Women physicians, substance abuse, relationship troubles
- Burnout - State of emotional and physical exhaustion caused by prolonged stress, relieved after relaxation
- Overworking - The feeling goes away as soon as you leave the hospital

Depression in doctors:

- Rates of depression are high in doctors.
- It is difficult to diagnose depression in doctors as they won't seek medical help because of the fear of being judged personally and professionally.
- Depression if untreated is a significant risk factor for suicide.

Suicide in doctors:

The risk of suicide is more among female physicians. Risk factors for suicide can be:

- Personal factors: Underlying psychiatric disorders, relationship troubles, stress, trauma, etc.
- Structural factors: It refers to the condition in which the doctors work. These include increased managerialism, paperwork, rushed patient interaction, inability to time off or rest, work pressure, etc.
- Endemic factors: Exposure to trauma and deaths.

Burnout:

It is classified in ICD 11 as an occupational phenomenon. It is defined as a syndrome resulting from chronic workspace stress that has not been managed successfully. It has three dimensions:

- having a feel of energy depletion or exhaustion
- Feeling of negativism or cynicism related to the job, or having increased mental distance from the job
- decreased professional efficacy.

It is a phenomenon related to occupational context and is not classified as a medical condition.

Solution to Question 3:

When a patient is using nicotine gum for nicotine substitution therapy, there should be a gap between nicotine gum and coffee/soda/acidic food as it decreases its absorption.

Treatment of nicotine addiction

1. Behavioral therapy- Elimination of the factors that favor smoking in the patient's environment, skill training, etc.

2. Pharmacotherapy used in the treatment of nicotine addiction are:

- Nicotine replacement therapy- It is used to reduce the symptoms of nicotine withdrawal without the use of tobacco. It is given for the initial maintenance period followed by a reduction period both lasting for about 6-12 weeks. It is available in various forms like

- Nicotine gum- It releases nicotine by chewing and buccal absorption. Smokers can use 1-2 gums per hour and a maximum of 24 gums per day. Acidic beverages like coffee, tea, soda, and juice should not be used during gum use because they decrease absorption. Poor compliance is a drawback.

- Nicotine lozenges- This is especially useful for people who smoke immediately after awakening. Users must suck the lozenges until it is completely dissolved and must not swallow it. It provides the highest level of nicotine compared to other nicotine replacement therapies.

- Nicotine patches- It provides continuous delivery of nicotine and is available as 16 or 24-hour tapering preparations. It has good compliance.

- Nicotine nasal spray- Provides nicotine concentration in the blood similar to smoking. Major adverse effects are rhinitis, watering eyes, and coughing

- Nicotine inhaler - Even though it was designed to deliver the nicotine to the lungs, the actual absorption occurs in the upper throat.

- Non-nicotine medications

- Bupropion is an antidepressant medication. It has dopaminergic and adrenergic actions.

- Varenicline is a partial agonist of alpha-4 beta-2 nicotinic receptor. It blocks nicotine from binding to the receptor and decreases the rewarding aspects of cigarette smoking. At the same time, it also stimulates the receptor and reduces withdrawal and cravings. There were concerns about its neuropsychiatric and cardiovascular side effects, but according to recent studies varenicline is considered a safe drug.

- Nortriptyline - Second-line drug

- Clonidine- It decreases the sympathetic activity from locus ceruleus and is useful in controlling the withdrawal symptoms.

Nicotine addiction:

It acts on the nicotinic subtype of acetylcholine receptors. Its addictive properties are thought to be due to the activation of the dopaminergic pathway in the ventral tegmental area. It causes improved attention, problem-solving ability, learning, and lifts the mood of the users.

Withdrawal symptoms include intense craving for tobacco, irritability, difficulty concentrating, drowsiness, altered sleep, decreased heart rate, and blood pressure, decreased motor performance, and increased muscle tension.

Solution to Question 4:

The correct option is A-2,B-4,C-1,D-3.

- Twilight state - Impaired level of consciousness
- Oneiroid state - Oneirophrenia
- Oligophrenia - Subnormal intelligence
- Paranoid ideation - Persecution

Twilight state - It refers to a state of disturbed consciousness along with hallucinations.

Oneiroid state or oneirophrenia - It refers to dream-like delusional derangement of consciousness in which the patient is not fully oriented in time and place. The characteristic feature of this state is the presence of kaleidoscopic quality of the dream-like experiences where reality, illusions, and hallucinations are merged into one. It is associated with catatonic disturbances. The term oneiroid schizophrenia refers to patients engaged in their hallucinations to an extent where there is the exclusion of involvement in the real world.

Oligophrenia - It refers to a group of mental states characterized by subnormal levels of intellectual activity. It is either congenital or acquired during early infancy. It is associated with other cognitive disturbances, affective disorders, and volitional disorders.

Paranoid ideation: It refers to thinking that is dominated by persecutory, suspicious, or grandiose contents. The thought content is less than that for delusions. Paranoid delusions refer to persecutory delusions and delusions of grandeur, control, and reference.

Solution to Question 5:

Post-traumatic stress disorder(PTSD) is not classified as OCD under ICD 11.

Obsessive-compulsive or related disorders as per ICD 11

It refers to a group of disorders that are characterized by repetitive thoughts and behaviors that are similar in etiology and important diagnostic validators. Important features of these disorders are obsessions, intrusive thoughts, and preoccupations. It includes:

- Obsessive-compulsive disorder
- Body dysmorphic disorder
- Hypochondriasis
- Olfactory reference disorder
- Hoarding disorder
- Body focused repetitive behavior disorders (Skin pricking, hair pulling - Trichotillomania)

Important points regarding ICD-11

The ICD 11 was released by the WHO in June 2018 and will come into effect on 1 January 2022.

Its designations for antibiotic resistance align with the Global Antimicrobial Resistance System (GLASS).

New chapters on

- Traditional medicine
- Sexual health: Gender incongruence has been moved from mental health to sexual health.
- Immune system disorders
- Diseases of the Blood and Blood forming Organs
- Sleep-Wake Disorders
- Extension codes

Gaming disorder has been added under Addictive disorders.

Solution to Question 6:

Obsessive-compulsive disorder (OCD) is commonly associated with depression. The lifetime prevalence of the major depressive disorder in patients with OCD is 67 percent. Other common psychiatric comorbidities in patients with OCD are social phobia, alcohol use disorders, panic, and personality disorders.

Obsessive-compulsive disorder (OCD) comprises symptoms that include intrusive thoughts, preoccupations, and compulsions. These recurrent obsessions or compulsions cause severe distress to the patient. The patient might have an obsession, a compulsion, or both.

An obsession is a thought, feeling, or sensation that is repetitive and intrusive. A compulsion is a conscious, recurrent behavior, eg: counting, checking, and are done to reduce the anxiety associated with the obsession. However, this act may not reduce the anxiety and might increase the anxiety even more.

Symptom patterns:

- Contamination - This is the most common symptom pattern of OCD. The patient feels that the objects are contaminated. They repeatedly wash their hands or avoid touching objects which are thought to be contaminated. In extreme cases, patients may rub the skin off their hands due to excessive hand washing or may refuse to leave the house in fear of germs.
- Pathological doubt - This is the second most common symptom pattern. The patient presents with an obsession with doubt and a compulsion to check. This obsession arises due to possible danger or violence (e.g., forgetting to turn off the stove or not locking a door) which is followed by multiple trips to check, for example, if the stove is turned off, if the door is locked, etc.
- Intrusive thoughts - This is the third most common symptom pattern. These patients have intrusive obsessional thoughts without compulsion. These thoughts could be sexual or an aggressive act that the patient doesn't agree. The patients might report themselves to the police or confess to a spiritual leader.
- Symmetry - This is the fourth most common pattern which can lead to a compulsion of slowness. Patients can take hours to eat a meal or shave their faces.

Diagnosis:

- Presence of obsessions, compulsions or both
- Symptoms are time-consuming
- Symptoms not related to substance abuse
- Not better explained by other mental disorders

Treatment of OCD includes:

- Pharmacotherapy: Selective serotonin reuptake inhibitors (SSRI) or clomipramine
- Behavioral therapy: Approaches include:
 - exposure and response prevention
 - desensitization
 - thought stopping
 - flooding
 - implosion therapy
 - aversive conditioning

A combination of behavioral and pharmacotherapy is most effective.

Other options

Option A - Atypical antipsychotics are not the first-line in the treatment of OCD. They can be given only when SSRIs and clomipramine (tricyclic antidepressant) are not effective.

Option B - The prevalence of OCD among the general population is 2-3 percent and not 7-10 percent.

Option D - Contamination is the most common obsession.

Solution to Question 7:

In Korsakoff's psychosis, there is anterograde amnesia with loss of recent memory.

Korsakoff's syndrome is a chronic amnestic syndrome that can follow Wernicke's syndrome. It is caused by thiamine deficiency that is associated with chronic alcohol abuse. The typical feature of this disease is anterograde amnesia with loss of recent memory in an alert and responsive patient. Patients with Korsakoff's syndrome also typically have a change in personality, with a lack of initiative, reduced spontaneity, and a lack of interest.

The treatment is with 100 mg of thiamine orally for 3-12 months. Despite adequate treatment and nutritional support, very few of the patients fully recover. With treatment, patients may remain amnestic for up to 3 months, and then improve gradually during the next year.

Prolonged use of alcohol causes alcohol-induced persisting amnestic disorder with loss of short-term memory. When the symptoms are acute, it is called Wernicke's encephalopathy and when chronic, it leads to Korsakoff's syndrome. These conditions are caused due to thiamine deficiency and when they occur together, they are called Wernicke-Korsakoff syndrome.

Other options

Option A -Triad of confusion, ataxia, and ophthalmoplegia is seen in Wernicke's encephalopathy. There may also be vestibular dysfunction. Wernicke's encephalopathy is completely reversible. The treatment is parenteral administration of 100 mg thiamine for 1-2 weeks.

Option B - Personality is affected in Korsakoff's syndrome. Recent memory is affected more compared to long-term memory.

Option D - Implicit memory refers to memory that does not require conscious recollection. Implicit aspects of memory are preserved, however, some studies conclude that implicit memory may be intact or impaired depending on the specific task parameters.

Psychiatry NEET 2021

Question 1:

During psychotherapy, the therapist had mixed conscious and unconscious feelings towards his patient. This is known as_____.

- a) Countertransference
- b) Transference
- c) Dissociation
- d) Preoccupation

Question 2:

A patient stopped alcohol consumption for 3 days and presented with irritability, disorientation, paranoid delusions, agitation, visual hallucinations, and altered sensorium. What is the likely diagnosis in this case?

- a) Delirium tremens
- b) Wernicke's encephalopathy
- c) Korsakoff psychosis
- d) Alcohol induced psychosis

Question 3:

A teacher demonstrated the steps of handwashing to her students. Students also performed the same activity later at school and at home. What is the type of learning in this case?

- a) Cognitive
- b) Psychomotor
- c) Affective
- d) Affective and cognitive

Question 4:

Which of the following is a formal thought disorder?

- a) Obsession
- b) Derailment
- c) Delusion
- d) Thought insertion

Question 5:

A 16-year-old girl presented with overfamiliarity, flight of ideas, elevated mood, increased sexual desire, and pseudohallucination. What is the likely diagnosis?

- a) Mania
- b) Hypomania
- c) Cyclothymia
- d) Schizomania

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	b
5	a

Detailed Explanations

Solution to Question 1:

In the given clinical scenario, the therapist has mixed conscious and unconscious feelings towards his patient. This is known as counter-transference.

Counter-transference is the positive and negative feelings/emotions expressed by healthcare professionals towards their patients. It is the process of the physician unconsciously displacing onto the patient patterns of behaviors or emotional reactions as if he/she were a significant figure from earlier in the physician's life. Psychiatrists should be alert to signs of countertransference issues like missed appointments by the psychiatrist, boredom, or sleepiness during a session. Supervision and personal therapy can help the psychiatrist recognize and deal with these issues.

Transference is the positive and negative feelings expressed by the patients towards the healthcare professional. It is the process of the patient unconsciously displacing onto individuals patterns of behavior and emotional reactions that originated with significant figures from the past (generally in childhood). This displacement is onto the psychiatrist in a clinical scenario, who is a parent surrogate. The patient may be angry, hostile, demanding, or obedient to an extreme degree because of former relationships and patterns of behaviors. Failure to recognize this can make the psychiatrist react to the patient's behavior like a personal attack.

These are not defense mechanisms but are commonly encountered during clinical practice or while giving psychotherapy. In an interview, transference and counter-transference are very significant expressions of unconscious processes.

Option C: Dissociation is an immature defense mechanism. In this, involuntary splitting/suppression of mental functions occurs from the rest of the personality in a way that allows the expression of forbidden unconscious impulses without having a sense of responsibility for actions.

Solution to Question 2:

The given clinical scenario of a patient who has not consumed alcohol for the past 3 days with disorientation, paranoid delusions, visual hallucinations, and altered sensorium is suggestive of delirium tremens.

Delirium tremens is a medical emergency that occurs during alcohol withdrawal. It is an acute condition and usually develops within 72 hours after the last drink. It has symptoms of both delirium and alcohol withdrawal. The clinical features of delirium tremens are:

- Altered sensorium with disorientation
- Perceptual distortions, most frequently tactile or visual hallucinations
- Autonomic hyperactivity with tachycardia, fever, anxiety, insomnia, hypertension. And sweating
- Fluctuating levels of psychomotor activity, ranging from agitation to lethargy
- Altered sleep-wake cycle

Other options:

Option B - Wernicke's encephalopathy (alcoholic encephalopathy) is associated with thiamine deficiency. Alcohol abuse may lead to thiamine deficiency because of poor nutritional habits, or malabsorption. It is an acute neurological disorder characterized by:

- Global confusion
- Ocular abnormalities like horizontal nystagmus, gaze palsy, anisocoria, and sluggish pupillary reflex
- Ataxia - mainly affecting the gait
- Vestibular dysfunction

It may resolve with thiamine therapy or progress to Korsakoff's syndrome.

Option C - Korsakoff's psychosis can follow Wernicke's encephalopathy due to thiamine deficiency. It is a chronic amnestic syndrome that is characterized by the following clinical symptoms in an intact and responsive patient:

- Impaired mental syndrome, especially recent memory
- Anterograde amnesia
- Confabulation

Option D - Alcohol-induced psychosis is characterized by the development of auditory hallucinations or paranoid delusions in the context of a heavy drinking or alcohol withdrawal. These patients have a clear sensorium, unlike the patients having delirium tremens.

Note: Tremulousness is the first alcohol withdrawal symptom.

Solution to Question 3:

The type of learning seen in this case is psychomotor learning.

The psychomotor domain includes learning by observation, physical movements, and the usage of motor skill areas of the brain. The development of these skills requires practice, depending upon the complexity of the task. It is measured based on speed, precision, distance, and procedures in execution.

The three domains in learning are the cognitive, psychomotor, and affective domains. They are described as follows.

Cognitive learning includes the domain of knowledge. Bloom's taxonomy described six levels of cognitive skills. This was revised to include types of knowledge and cognitive processes. The cognitive processes include remembering, understanding, applying, analyzing, evaluating, and creating. Eg. Solving multiple-choice questions in the NEET PG entrance exam involves the usage of the cognitive domain.

Psychomotor learning includes the domain of skills. It includes imitation, manipulation, precision, articulation, and naturalization. Eg. Learning how to insert an IV cannula by observing the residents and nurses during the internship involves the usage of the psychomotor domain.

Affective learning includes the domain of attitude and values. It includes receiving, responding, valuing, organizing, and characterization by value. Eg. Communicating with patients with empathy, responding to their concerns, and interacting with the relatives of patients involves the usage of the affective domain.

Integration of all these three methods in teaching and learning methodology leads to a better understanding of the subject.

Solution to Question 4:

Derailment is a formal thought disorder.

It is a breakdown in both the logical connection between ideas and the overall sense of being goal-oriented. It is generally identified in schizophrenic patients as a generalized loosening of associations.

For example, a person with derailment thought disorder might jump from talking about rabbits to the hair on their head to your sweater. They talk in chains of semi-related ideas. Their communication style has a generalized loosening of associations. Their ideas often fall further and further from the topic.

Formal thought disorders are:

- Circumstantiality is the overinclusion of trivial or irrelevant details that impedes the sense of getting to the point.
- Clang association is an association of thoughts with the sound of words rather than by their meaning (e.g., through rhyming).
- Derailment is the breakdown in both the logical connection between ideas and the overall sense of goal-directedness. The words make sentences, but the sentences do not make sense.
- Flight of ideas is a succession of multiple associations so that thoughts seem to move abruptly from idea to idea, through rapid, pressured speech.
- Neologism is the invention of new words or phrases or the use of conventional words in idiosyncratic ways.
- Perseveration is the repetition of out-of-context words, phrases, or ideas.
- Tangentiality is responding to a question, that is appropriate to the general topic without actually answering the question.
- Thought blocking is a sudden disruption of thought or a break in the flow of ideas.

Other options:

Option A - An obsession is a recurrent and intrusive thought, feeling, idea, or sensation.

Option C - A delusion is defined as a false belief based on incorrect inference about external reality that firmly sustains despite evidence to the contrary.

Option D - Thought insertion is a type of delusion where the patient believes that thoughts are not his/her own and have been inserted into his/her mind.

Solution to Question 5:

In the given clinical scenario, history of overfamiliarity, elevated mood, flight of ideas, and increased sexual desire are suggestive of mania.

The diagnostic criteria for mania are:

A. Distinct period of abnormally and persistently elevated/irritable mood, and abnormally and persistently increased activity/energy, lasting at least 1 week.

B. During the period of mood disturbance and increased energy/activity, three (or more) of the following symptoms are present to a significant degree:

1. Increased self-esteem or grandiosity

2. Decreased need for sleep
3. More talkative than usual or a pressure to keep talking
4. Flight of ideas
5. Distractibility
6. Increase in goal-directed activity or psychomotor agitation
7. Excessive involvement in activities that have a high potential for painful consequences (unrestrained buying sprees, sexual indiscretions)

C. The mood disturbance is severe enough to cause marked social or occupational impairment or to necessitate hospitalization.

D. The episode is not attributable to the physiological effects of a substance (e.g., a drug of abuse, a medication, other treatment) or to another medical condition.

Other options:

Option B - Hypomania is a distinct period of abnormally and persistently elevated/irritable mood, abnormally and persistently increased activity/energy, lasting for at least 4 days. It is similar to a manic episode but it is not severe enough to cause impairment in social or occupational functioning. No psychotic features are present.

Option C - Cyclothymia is a mild, chronic mood disorder with numerous depressive and hypomanic episodes over the course of at least 2 years.

Option D - Schizomania is an old term in DSM III, used to describe schizophrenic and manic symptomatology occurring in one episode, classified under schizoaffective disorder.

Schizoaffective disorder is characterized by active phase symptoms of schizophrenia, like delusions, hallucinations, disorganized speech, grossly disorganized behavior, or negative symptoms. Additionally, there is a presence of manic/depressive syndrome that is not brief relative to the duration of the psychosis. The delusions or hallucinations are present for at least 2 weeks without coexisting mood symptoms.

Psychiatry INI-CET 2022

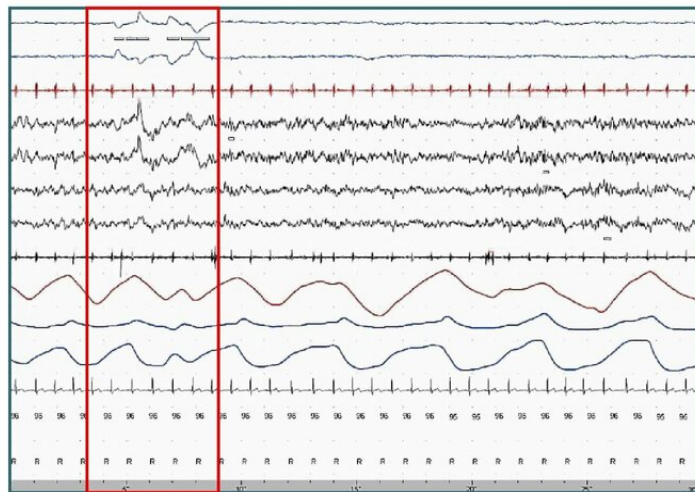
Question 1:

Which of the following is false about bipolar disorder?

- a) It most commonly affects ages 35-45 years
- b) Type 1 is common in both sexes
- c) Prevalence in general population is 1-2%
- d) Attempt to commit suicide is seen around 15% of people

Question 2:

On performing polysomnography in a patient, the waves of EOG, EEG, and EMG respectively from above downwards are seen below. Which stage of sleep does the marked area represent?



- a) REM sleep
- b) NREM I sleep
- c) NREM II sleep
- d) NREM III sleep

Question 3:

A patient diagnosed with schizophrenia was started on risperidone. After taking the medication last night, the patient rushes to OPD today with complaints of upwards rolling of his eyes. What is the management of the condition?

- a) Lorazepam
- b) Promethazine
- c) Olanzapine
- d) Stop antipsychotic

Question 4:

Which of the following disease is exclusively seen in females?

- a) Rett syndrome
- b) Asperger syndrome
- c) Selective mutism
- d) Atypical autism

Question 5:

Match the following:

- a) 1-D, 2- B, 3-A, 4-C
- b) 1-C, 2-B, 3-D, 4-A
- c) 1-C, 2-D, 3-B, 4-A
- d) 1-D, 2-C, 3-B, 4-A

Question 6:

A 14-year-old girl presented with sudden-onset blindness for the past 4 hours. However, on history taking, it is noted that she is not concerned about it. However, she is concerned that her mother passed away recently and that she should have spent more time with her. Physical examination findings are normal. Which of the following is true about the condition?

- a) In children, equally among males and females.
- b) In adults, equally among males and females.
- c) In adults, occurs more in males than in females.
- d) In children, occurs more in females than in males.

Question 7:

What is the dose range of lithium in mEq/L for acute attacks of mania and long-term maintenance?

- a) 0.5 to 1 mEq/L
- b) 0.1 to 0.5 mEq/L
- c) 0.6 to 1.5 mEq/L
- d) 2 to 3 mEq/L

Question 8:

A patient who is a chronic alcoholic presents with seizures and altered sensorium. He has been drinking for the past 10 years and takes 24 units per day. His last drink was 2 days back. He is found to have auditory hallucinations and secondary delusions. What will you use for the management of this patient?

- a) 1,4
- b) 1,2,4
- c) 1,3,4
- d) 1,2

Question 9:

Match the following:

- a) 1-B;2-A;3-D;4-C
- b) 1-C;2-A;3-D;4-B
- c) 1-C;2-D;3-A;4-B
- d) 1-C;2-B;3-D;4-A

Question 10:

Which of the following is the most appropriate definition of stress?

- a) External pressure
- b) Mental tension

- c) It is the body's response to anything that requires attention or action causing physical, emotional, or psychological strain.
- d) Internal strain

Question 11:

A pregnant mother was on SSRIs. The risk of which of the following is increased in the newborn?

- a) A, C
- b) A, B, C, D
- c) B, C, D
- d) A, B, C

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	a
5	c
6	d
7	c
8	d
9	b
10	c
11	b

Detailed Explanations

Solution to Question 1:

The false statement among the following is option A. The age of onset for bipolar type I disorder ranges from, children of age 5 or 6 years to adults of age 50 years or older, with a mean age of 18

years. The mean age of onset for bipolar type II is 20 years.

Patients with bipolar disorder can present with both manic and depressive episodes or with manic episodes alone. The bipolar disorder presented with mania and depression for at least a week is known as bipolar type 1 disorder. Whereas, if it is presented with hypomania and depression it is known as bipolar type 2 disorder.

The lifetime prevalence of bipolar type 1 disorder is 0 to 2.4%, whereas bipolar type 2 disorder is 0.3-4.8%. Bipolar type 1 disorder has an equal prevalence among men and women. Suicide attempt is seen in around 15% of people with bipolar disorder.

A manic episode is defined as a persistent elevated, expansive, or irritable mood for at least 1 week which requires hospitalization. A hypomanic episode is also similar to a manic episode except it doesn't cause social and occupational functioning impairment. It lasts for at least 4 days and doesn't show psychotic symptoms.

A person is said to be depressed if he has at least four of the following symptoms for at least 2 weeks:

- Change in appetite and weight
- Change in sleep and activity
- Lack of energy
- Guilty feeling
- recurring thoughts about death or suicide.

Lithium is used for the pharmacological management of bipolar disorder.

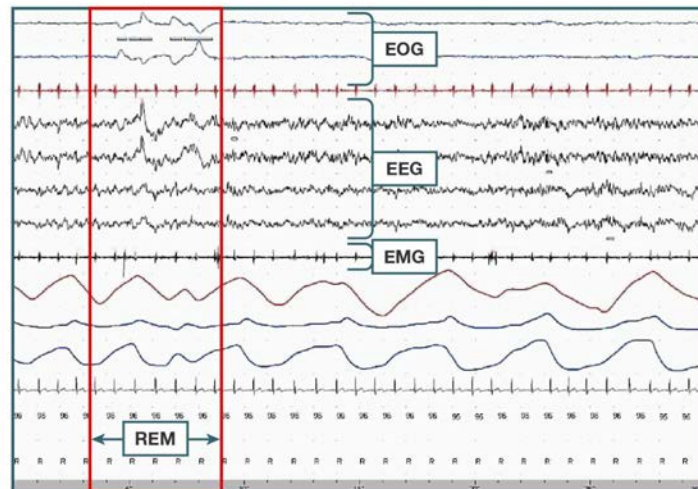
Solution to Question 2:

REM sleep is represented by the marked area in the given polysomnography.

The three zones in the given image represent EOG, EEG, and EMG, respectively. In the marked area,

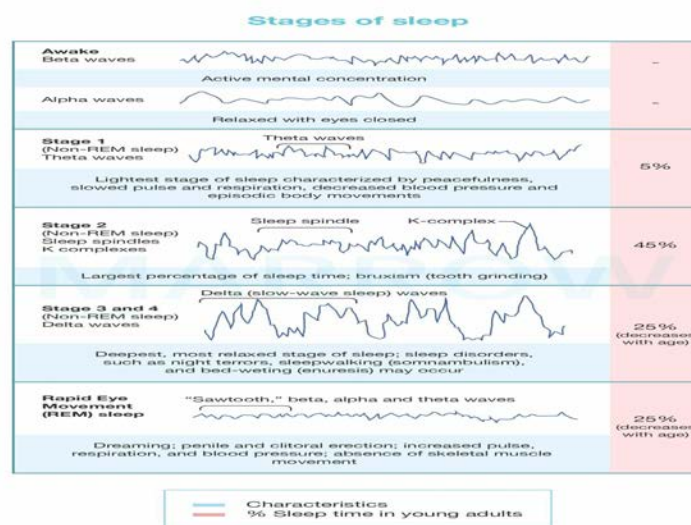
- EOG shows Positive activity
- EEG shows Mixed waves
- EMG shows Minimal activity

This is characteristic of REM sleep.



The following table helps in identifying the state of wakefulness and sleep from the polysomnograph :

	EEG	EOG	EMG
Active state	Fast	+	+
REM sleep	Mixed	+	-
NREM sleep	Slow	-	+



Solution to Question 3:

The given clinical scenario is suggestive of oculogyric crisis and promethazine is used for the management.

Oculogyric crisis refers to the spasm of the extraocular muscles, usually causing upward and outward deviation of the eyes. It is a severe form of acute dystonia, an extrapyramidal side effect of antipsychotics like risperidone. It can be treated by drugs with antimuscarinic activities like promethazine.

Dystonia is acute in onset and occurs within 1-5 days of starting treatment. It is more common in the younger population. The proposed mechanism of action is D2 blockade in the nigrostriatal pathway. There is typical involvement of head and neck muscles and presents in a variety of forms, such as torticollis, trismus, mouth opening dystonia, grimacing, dysarthria, oculogyric crisis, blepharospasm, and swallowing difficulties.

Risperidone is an atypical antipsychotic with D2 blocking but strong 5HT₂ antagonistic activity. Its main side effects include weight gain, hyperglycemia, hyperprolactinemia, dyslipidemia, fatigue, etc. At low doses, extrapyramidal side effects are less, but can be significant in higher doses. Extrapyramidal side effects include;

- Parkinsonism: Features are bradykinesia, rigidity, variable tremor, masked facies, and shuffling gait.
- Acute dystonia: Which causes spasms of muscles of the tongue, face, eyes, neck, back, etc.
- Neuroleptic malignant syndrome: Extreme rigidity, fever, unstable blood pressure, myoglobinemia, etc.
- Rabbit syndrome: Perioral tremors occur as a delayed variant of parkinsonism.

Other options:

Option A: Lorazepam is a benzodiazepine. Its adverse effects include sedation, light-headedness, confusional state, weight gain, etc.

Option C: Olanzapine is also an atypical antipsychotic. Its adverse effects are dry mouth, constipation, weight gain, impaired glucose tolerance etc.

Solution to Question 4:

Rett syndrome is exclusively seen in females. It is caused by X-linked dominant inheritance and hence affects only females, as most males die in utero.

Rett syndrome is a form of autism spectrum disorder. The child is normal at birth, but by 1 year of age deterioration in communication skills, motor behaviour, and social functioning are noticed. Symptoms are ataxia, loss of speech, teeth grinding, and facial grimacing. When the child is awake, intermittent hyperventilation and disorganized breathing pattern are observed. Stereotypical hand movements like hand-wringing, are characteristically seen. Progressive gait disturbance and seizures are seen. Decreased pigmentation of the substantia nigra, along with cerebral atrophy occurs, implying an abnormal dopaminergic nigrostriatal system.

Other options:

Option B: Asperger syndrome is a version of autism spectrum disorder. It is characterized by limited social interaction and restricted interest and behaviour. There is no delay in language

development.

Option C: Selective mutism is a social anxiety disorder. The child hesitates to speak in one or more social settings, mostly the school.

Option D: Atypical autism is autism, atypical either in the age of onset or in symptomology or both.

Solution to Question 5:

The correct option is 1-C, 2-D, 3-B, 4-A

Impulse control disorders are characterized by the repeated failure to resist an impulse, drive, or urge to perform an act that is rewarding to the person, at least in the short-term, despite consequences such as long-term harm either to the individual or to others; marked distress about the behavior pattern; or significant impairment in personal, family, social, educational, occupational, or other important areas of functioning.

It includes

- Pyromania
- Kleptomania
- Compulsive sexual behavior disorder
- Intermittent explosive disorder

Kleptomania is the repeated failure to resist impulses to steal objects that are of no value to the person. The individual can usually afford the object or does not need it, but it is the act of stealing itself that is the goal. The stolen objects are usually given away, returned, or kept hidden. The act is not planned and is carried out alone.

Pyromania is the deliberate setting of fires on more than one occasion.

Pyromania is significantly associated with the following:

- Substance abuse disorder (especially alcoholism).
- Affective disorders (depression or bipolar)
- Other impulse control disorders, such as kleptomania in female firesetters.
- Personality disturbances, such as borderline personality disorders.
- Attention-deficit/hyperactivity disorder and learning disabilities in childhood pyromania.

Dipsomania is an irresistible impulse to drink alcohol at a periodic intervals.

Mutilomania is an irresistible impulse to mutilate animals.

Solution to Question 6:

The above scenario is suggestive of conversion disorder. Option D is a true statement regarding the above condition as it is more commonly seen in females.

Conversion disorder is also called functional neurological symptom disorder, and there is no bodily cause involved in this condition. A conversion disorder is an acute and temporary loss or alteration in motor or sensory function triggered by psychosocial stress. Most of the patients have a history of traumatic events in childhood. The symptoms or deficits of conversion disorder are not intentionally produced but are a symbolic representation of an unconscious conflict (primary gain). It is more commonly seen in females.

Primary gain refers to keeping internal conflicts outside of awareness and the relief from stress. Secondary gain refers to the attention gained by the patient (from bystanders) as a result of being sick. If the secondary gain is intentional, then it would lead to the diagnosis of malingering. A related phenomenon is la belle indifference, in which the patient is not concerned about the serious symptoms.

Symptoms of conversion disorder:

- Mutism, blindness, and paralysis are the most common symptoms.
- Sensory system: Anesthesia, paresthesia, deafness, blindness, and tunnel vision
- Motor system: Abnormal movements, gait disturbance (astasia abasia), weakness, paralysis, tics, jerks, etc.
- CNS: Pseudo-seizures

Treatment: Hypnosis, anxiolytics, and behavioral relaxation exercises are effective. Psychoanalysis and insight-oriented psychotherapy help in the resolution of the symptoms.

Solution to Question 7:

Dose range of lithium in mEq/L for acute attacks of mania and long-term maintenance is 0.6 to 1.5 mEq/L

Lithium has a narrow therapeutic index (0.8 to 1.2 mEq/L) and, therefore, requires monitoring to suspect toxicity. The toxic effects of lithium usually start manifesting when the serum level crosses 1.5 mEq/L.

The range of 1.0-1.5 mEq/L is only favoured for acute attacks of mania with lower doses of 0.6-1.0 mEq/L for long term maintenance.

Acute intoxication is characterized by

- Vomiting
- Profuse diarrhea
- Coarse tremors
- Ataxia
- Coma
- Convulsions.

A fine action tremor is common in patients taking lithium at therapeutic doses and can develop at any time during treatment. Therapeutic lithium tremor is generally symmetric, limited to the arms, and nonprogressive. The tremor from lithium toxicity is coarser and more disabling, may

also affect the legs, and occurs with other symptoms of toxicity.

Serious effects of lithium toxicity will involve confusion, hyperreflexia, and seizures. Motor and cognitive adverse effects can become irreversible, with persistent cerebellar tremor being the most common.

Solution to Question 8:

The given clinical scenario is suggestive of delirium tremens. Management of this patient involves the administration of diazepam and thiamine.

Delirium tremens is the most severe form of alcohol withdrawal syndrome. It occurs usually within 3–4 days of alcohol withdrawal.

Characteristic features of delirium tremens:

- Clouding of consciousness, with disorientation to time and place
- Poor attention and distractibility
- Visual hallucinations and illusions, which are often vivid and very frightening
- Autonomic disturbances like tachycardia, fever, hypertension, diaphoresis, and pupillary dilatation
- Psychomotor agitation and ataxia

Benzodiazepines such as diazepam (intravenously) or chlordiazepoxide (orally every 4 hours) are given. It is also recommended to administer thiamine in patients with alcohol-related disorders to prevent Wernicke-Korsakoff syndrome.

Acute alcohol intoxication can be treated with the evacuation of stomach and bowel and gastric lavage with an alkaline solution. Administration of 1 L of normal saline with 10% glucose, 100 mg thiamine, and 15 U insulin may be useful. If the coma progresses deeper, stimulants like strychnine and caffeine can be used. Patients should be started on artificial respiration if there is respiratory depression or difficulty breathing.

Chronic alcohol intoxication can be treated with disulfiram (daily dose of 250 mg OD), conditioned reflex treatment, hypnosis, and psychotherapy.

According to the Motor Vehicle Amendment bill (2019), whoever while driving, or attempting to drive, a motor vehicle, has, in his/her blood, alcohol exceeding 30 mg/100 mL of blood, detected in a test by a breath analyzer, is punishable by law.

Naltrexone is an opioid receptor antagonist which is used along with cognitive behavioral therapy in cases of opioid withdrawal and alcohol withdrawal. It reduces craving and consumption, also reduces the severity of withdrawal symptoms.

Flupenthixol is an antipsychotic antagonist of both D1 and D2 dopamine receptors. The depot injections are used for the management of schizophrenia and depression.

Solution to Question 9:

The correct answer is option B: 1-C;2-A;3-D;4-B

Paraphilias or perversions are sexual stimuli or acts that are deviations from normal sexual behaviors but are necessary for some people to experience arousal and orgasm.

Eonism (also known as transvestism) is a sexual perversion, where a person's whole personality is dominated by a desire to be identified as a member of the opposite sex. This is usually seen in males who obtain sexual pleasure by wearing female clothes. There is no hormonal/genital abnormality.

Exhibitionism is the urge to expose genitals to a stranger or unsuspecting person.

Frotteurism refers to the sexual gratification achieved by touching or rubbing against a non-consenting person. This behavior often occurs in busy and crowded places. It is predominantly described in men.

Necrophilia is an obsession with obtaining sexual gratification from dead bodies.

1. Eonism	C. Sexual pleasure by wearing clothes of the opposite sex
2. Exhibitionism	A. Exposure of one's genitals to an unsuspecting stranger
3. Frotteurism	D. Sexual gratification by rubbing private parts against another person
4. Necrophilia	B. Sexual gratification by intercourse with dead bodies

Solution to Question 10:

According to the WHO, stress can be defined as any type of change that causes physical, emotional, or psychological strain. Stress is your body's response to anything that requires attention or action. Everyone experiences stress to some degree.

Solution to Question 11:

The newborn of a pregnant mother on SSRI has an increased risk for A, B, C and D. SSRIs like sertraline are the most commonly used antidepressants to treat depression in pregnant women.

SSRIs and maternal depression have both been found to cause a low APGAR score.

The development of ADHD in the newborn has been found to be commonly associated with untreated maternal depression. But studies regarding SSRIs having a causal association with the development of ADHD have been inconclusive. This can be contributed to maternal depression being a confounder.

When used in the second half of pregnancy, SSRIs have been associated with an increased risk of precipitating persistent pulmonary hypertension in the newborn.

There has been evidence of a transient delay in developmental milestones in patients who were exposed to SSRIs during pregnancy. Most studies have observed that the development returns to normal after some time. But if the mother is severely depressed, delayed developmental milestones have also been attributed to the mother's disinterest in the infant.

The other side effects of SSRIs include:

System	Side effects of SSRIs
Sexual	Anorgasmia, Decreased libido
Gastrointestinal (5-HT ₃ receptor)	Nausea, diarrhea, anorexia, vomiting, flatulence, and dyspepsia
Nervous system	Extremely vivid dreams or nightmares, Anxiety, Insomnia/sedation, seizures, emotional blunting
Anticholinergic	Dry mouth, constipation, sedation
Cardiovascular	Drug-induced long QT syndrome
Hematological	Easy bruising
Endocrine	Mammoplasia, galactorrhea

Psychiatry NEET 2022

Question 1:

A 16-year-old girl has intense cravings for food. She eats large amounts of food, which is followed by self-induced vomiting. What is the probable diagnosis?

- a) Anorexia nervosa
- b) Bulimia nervosa
- c) Atypical depression
- d) Binge eating disorder

Question 2:

A woman, who is 4 days postpartum, presented with tearfulness, mood swings, and occasional insomnia. What is the likely diagnosis?

- a) Postpartum depression
- b) Postpartum blues
- c) Postpartum psychosis
- d) Postpartum anxiety

Answer Key

Question No.	Correct Option
1	b
2	b

Detailed Explanations

Solution to Question 1:

The probable diagnosis in the given scenario of a 16-year-old girl with intense food cravings followed by self-induced vomiting is bulimia nervosa.

Bulimia nervosa is an eating disorder, often seen in young women. It is characterized by binge-eating episodes, followed by inappropriate, compensatory measures to stop weight gain, like self-induced vomiting, laxative abuse, diuretics, enema, and emetic abuse. These binge-eating episodes occur at least once a week and are present for at least 3 months. The patient fears gaining weight and has a relentless drive to be thin.

The weight is normal in most patients with bulimia nervosa. However, few patients may either be overweight or underweight. There is a risk of dehydration and electrolyte imbalance.

Treatment modalities are cognitive behavioral therapy, psychotherapy, and antidepressants such as fluoxetine.

Other options:

Option A: Anorexia nervosa is an eating disorder that is often seen between the ages of 10 and 30. Typical features are self-induced starvation, voluntary and unhealthy reduction in weight, intense drive for thinness, and fear of becoming obese. These symptoms are usually present for at least a period of 3 months. Associated features can include depression, anxiety, amenorrhea in females, hypothermia, and bradycardia.

Option C: Atypical depression is a type of depression characterized by mood reactivity, which is brightening of the mood in response to positive events; significant weight gain or increase in appetite; hypersomnia; leaden paralysis; and/or interpersonal rejection sensitivity, resulting in significant social or occupational impairment.

Option D: Binge-eating disorder is the most common eating disorder. It is characterized by rapid intake of abnormally large quantities of food over a short period of time, even when not hungry. There are feelings of guilt about the eating episodes, but these patients do not compensate in any way following the episodes of binge.

Solution to Question 2:

Postpartum blues is the most likely diagnosis in a postpartum woman with tearfulness, mood swings, and insomnia from day 4.

Postpartum blues, also known as baby blues, is a transient mood disorder characterized by sadness, tearfulness, confusion, and dysphoria. There are no suicidal thoughts or thoughts of harming the baby. These symptoms typically develop 3-5 days after delivery. They are thought to be due to hormonal changes, the stress of giving birth, and the responsibilities of motherhood. They resolve spontaneously within 10-14 days.

Management involves reassurance, education, and support. The patient must be followed up and assessed if symptoms persist for >2 weeks to evaluate postpartum depression.

Other options:

Option A: When the symptoms of depressed mood, anxiety, insomnia, and weight changes start within 4 weeks as per DSM 5 or 6 weeks as per ICD 11, and the symptoms persist for ≥ 2 weeks, the diagnosis will be postpartum depression. The patient may have occasional thoughts of harming the baby and suicidal thoughts. It can persist for months to years when untreated.

Option C: Postpartum psychosis is a psychiatric emergency that usually occurs within 4 weeks after delivery (DSM 5) or 6 weeks after delivery (ICD 11). The predominant symptoms are

hallucinations, delusions, suicidal thoughts, and thoughts of harming the baby. Psychotic symptoms such as insomnia, restlessness, labile mood, and agitation are seen. Lithium, antipsychotics, and an antidepressant is the treatment of choice.

Option D: Postpartum anxiety is not defined by DSM-5. However, some platforms might describe postpartum anxiety as excessive or severe worry after childbirth.

Note: Zuranolone is the first-ever oral drug to be approved specifically for the treatment of postpartum depression. It is a neuroactive steroid that enhances the action of GABA in the brain and regulates mood.

Psychiatry NEET 2023

Question 1:

A schizophrenic patient was prescribed drug A after he did not respond to haloperidol and thioridazine. He now presents with excessive salivation, an increase in blood glucose, and hyperlipidemia. What is drug A?

- a) Ziprasidone
- b) Risperidone
- c) Clozapine
- d) Aripiprazole

Question 2:

A woman, who is 4 days postpartum, presented with tearfulness, mood swings, and occasional insomnia. What is the likely diagnosis?

- a) Postpartum depression
- b) Postpartum blues
- c) Postpartum psychosis
- d) Postpartum anxiety

Question 3:

A 25-year-old female reported symptoms of anxiety, palpitations, sweating, breathlessness, chest pain, and a feeling of impending doom. She recalls having experienced 5-6 similar episodes in the past 6 months, each episode lasting for about 20-30 minutes. What is the likely diagnosis?

- a) Depression
- b) Panic disorder
- c) Generalized anxiety disorder
- d) Phobia

Question 4:

A chronic alcoholic is brought to the emergency department with confusion, ataxia, and painful eye movements. The 6th cranial nerve is also involved. What is the likely diagnosis?

- a) Wernicke's encephalopathy
- b) Korsakoff psychosis
- c) Delirium tremens
- d) De Clerambault syndrome

Question 5:

A middle-aged man complains of very early ejaculation during intercourse. What is the non-pharmacological management that can be advised to the patient?

- a) Cognitive behavioral therapy
- b) Exposure and response prevention therapy
- c) Squeeze technique
- d) Sensate focus therapy

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	a
5	c

Detailed Explanations

Solution to Question 1:

Drug A is clozapine, an atypical antipsychotic used in the treatment of refractory schizophrenia. Excessive salivation, an increase in blood glucose, and hyperlipidemia are side effects of clozapine.

The side effects of clozapine are

- Agranulocytosis
- Weight gain
- Seizures
- Drooling
- Hyperthermia
- Hyperglycemia
- Hypertriglyceridemia

Clozapine causes the highest weight gain among all antipsychotics. Agranulocytosis is a dose-independent side effect and seizures are a dose-dependent side effect.

Other options:

Option A: Ziprasidone - Although it is an atypical antipsychotic, it does not typically cause excessive salivation, significant increases in blood glucose, or hyperlipidemia. The side effects of ziprasidone include orthostatic hypotension, QT prolongation, and minimal weight gain.

Option B: Risperidone - Risperidone is another atypical antipsychotic, but its side effect profile differs from the symptoms presented in the question. Risperidone is more likely to cause hyperprolactinemia, weight gain, sedation, dystonia, and parkinsonism features than the symptoms mentioned.

Option D: Aripiprazole - It is an atypical antipsychotic used to treat schizophrenia and bipolar disorder, but its side effects are typically less severe than those of clozapine. Aripiprazole is more likely to cause nausea, anxiety, and insomnia than the symptoms described.

Solution to Question 2:

Postpartum blues is the most likely diagnosis in a postpartum woman with tearfulness, mood swings, and insomnia from day 4.

Postpartum blues, also known as baby blues, is a transient mood disorder characterized by sadness, tearfulness, confusion, and dysphoria. There are no suicidal thoughts or thoughts of harming the baby. These symptoms typically develop 3-5 days after delivery. They are thought to be due to hormonal changes, the stress of giving birth, and the responsibilities of motherhood. They resolve spontaneously within 10-14 days.

Management involves reassurance, education, and support. The patient must be followed up and assessed if symptoms persist for ≥ 2 weeks to evaluate postpartum depression.

Other options:

Option A: When the symptoms of depressed mood, anxiety, insomnia, and weight changes start within 4 weeks as per DSM 5 or 6 weeks as per ICD 11, and the symptoms persist for ≥ 2 weeks, the diagnosis will be postpartum depression. The patient may have occasional thoughts of harming the baby and suicidal thoughts. It can persist for months to years when untreated.

Option C: Postpartum psychosis is a psychiatric emergency that usually occurs within 4 weeks after delivery (DSM 5) or 6 weeks after delivery (ICD 11). The predominant symptoms are

hallucinations, delusions, suicidal thoughts, and thoughts of harming the baby. Psychotic symptoms such as insomnia, restlessness, labile mood, and agitation are seen. Lithium, antipsychotics, and an antidepressant is the treatment of choice.

Option D: Postpartum anxiety is not defined by DSM-5. However, some platforms might describe postpartum anxiety as excessive or severe worry after childbirth.

Note: Zuranolone is the first-ever oral drug to be approved specifically for the treatment of postpartum depression. It is a neuroactive steroid that enhances the action of GABA in the brain and regulates mood.

Solution to Question 3:

The given clinical scenario is suggestive of a panic disorder. Panic disorder refers to an acute intense attack of anxiety accompanied by feelings of impending doom. Such attacks may not only occur in response to a fearful situation, or phobic objects, or in response to obsession or separation as in other anxiety disorders.

Panic disorder is characterized by the presence of an abrupt surge of intense fear or intense discomfort that reaches a peak within minutes. During the episode, the person may experience symptoms like palpitations, sweating, trembling or shaking, shortness of breath, nausea, dizziness, chest pain or discomfort or fear of dying. The patient also might feel like losing control, having a heart attack, or going crazy.

There is a dysfunction in major neurotransmitter systems like norepinephrine, serotonin, and GABA. Panic-inducing substances or panicogens may induce such attacks and include substances like 5-35% carbon dioxide, sodium lactate, bicarbonate, yohimbine (α blocker), flumazenil (GABA antagonist), cholecystokinin and caffeine.

It is not attributable to any drug abuse, medication, medical condition or mental disorder.

Treatment of this condition can be done by cognitive behavioral therapy and drugs like benzodiazepines (alprazolam, lorazepam), selective serotonin reuptake inhibitors (paroxetine), tricyclic antidepressants and monoamine oxidase inhibitors.

Solution to Question 4:

The above clinical history is suggestive of Wernicke's encephalopathy caused by thiamine (vitamin B₁) deficiency.

Wernicke's encephalopathy is an acute, neuropsychiatric syndrome characterized by nystagmus and ophthalmoplegia, mental status changes (confusion), and unsteadiness of stance and gait (ataxia). It is a medical emergency, and thiamine should be initiated immediately, either intravenously or intramuscularly. The current standard of treatment for patients with symptoms of Wernicke's encephalopathy is to give thiamine 100 mg intravenously before administering glucose-containing IV fluids.

Other options:

Option B: Korsakoff psychosis - While Korsakoff psychosis is also associated with thiamine deficiency and can occur along with Wernicke's encephalopathy, it is primarily characterized by memory loss and confabulation. The presence of confusion, ataxia, and painful eye movements with 6th cranial nerve involvement is more indicative of Wernicke's encephalopathy.

Option C: Delirium tremens - It is a severe form of alcohol withdrawal syndrome and usually occurs within 3-4 days of alcohol withdrawal. The features of delirium tremens are clouding of consciousness, disorientation to time and place, poor attention and distractibility, visual hallucinations and illusions, tachycardia, fever, hypertension, diaphoresis, and pupillary dilatation, and psychomotor agitation and ataxia.

Option D: De Clerembault syndrome (erotomania) - It is characterized by the delusional belief that someone is deeply in love with them. It is more common in women than in men, commonly a young woman who believes that a man whom she considers to be of higher social and/or professional class, is in love with her. This condition is not related to alcohol abuse and does not cause the neurological symptoms described in the question.

Solution to Question 5:

The clinical scenario of ejaculation very early on vaginal penetration is suggestive of premature ejaculation (PE), and the squeeze technique is a non-pharmacological management (behavioral method) used to prevent it.

PE is a disorder of the orgasm phase of the normal sexual cycle.

In premature ejaculation, there is a persistent or recurrent achievement of orgasm and ejaculation before the patient actually wishes to. According to DSM-5, it is classified as follows:

- Mild if ejaculation occurs within approximately 30 seconds to 1 minute of vaginal penetration
- Moderate if ejaculation occurs within approximately 15 to 30 seconds of vaginal penetration
- Severe if ejaculation occurs at the start of sexual activity or within approximately 15 seconds of vaginal penetration.

Pharmacologic therapy when combined with behavioral treatment appears to be more effective in treating premature ejaculation.

The squeeze technique is a behavioral method used on the penis to prevent premature ejaculation. Here, the penis is stimulated until just before orgasm after which it is gently squeezed.

Other options:

Option A: Although Cognitive behavioral therapy (CBT) can be beneficial for some patients with sexual dysfunction. It is not a specific technique for managing PE. Instead, CBT is more useful in addressing underlying psychological issues or relationship problems that may contribute to the condition. CBT is used in the treatment of depression, phobia, anxiety and panic disorders, OCD, personality disorders, and somatoform disorders.

Option B: Exposure and response prevention therapy is a strategy within CBT that is effective in the management of OCD. It is also useful in other conditions like tics but is not a suitable method for managing premature ejaculation.

Option D: Sensate focus therapy is a couple-oriented sex therapy that involves the exchange of sensual touch between partners to alleviate anxiety and negativity associated with sex and to build better sexual intimacy and communication. It does not specifically address the issue of premature ejaculation like the squeeze technique does and is hence not useful.

Psychiatry INI-CET 2023

Question 1:

A boy is experiencing low mood, decreased energy, and decreased interest in his surroundings. He is not taking food for the past one week. The thought sample says "I lost my head. I am dead. How can I eat?". What is the delusion seen in this case?

- a) Cotard's syndrome
- b) Hypochondriacal delusion
- c) Nihilistic delusion
- d) Bizarre delusion

Question 2:

La belle indifference is seen in:

- a) Mania
- b) Depression
- c) Schizophrenia
- d) Conversion disorder

Question 3:

SCOFF protocol is used for assessing:

- a) Mental health disorder
- b) Behavioural disorder
- c) Sexual health disorder
- d) Eating disorder

Question 4:

The tactile hallucinations seen in cases of chronic cocaine intoxication is?

- a) Oneroid state
- b) Formication

- c) Dipsomania
- d) Carphologia

Question 5:

Which of the following is used to define dependence?

- a) 1, 2
- b) 1 only
- c) 1, 2, 3, 4
- d) 1, 2, 3

Question 6:

A pregnant female with bipolar disorder came to the clinic in the third trimester. She has had 4 episodes in the past and is stable on lithium 750 mg/day since 1 year during which time she has not had any episodes. What is the next best step in management?

- a) Stop lithium and let her be drug free
- b) Increase lithium dosage
- c) Decrease lithium dosage
- d) Switch to valproate

Question 7:

HEADSSS assessment will be used by an intern for which of the following patient groups?

- a) Children under 5 years
- b) Neonates
- c) Infants
- d) Adolescents

Answer Key

Question No.	Correct Option
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1	c
2	d
3	d
4	b
5	c
6	c
7	d

Detailed Explanations

Solution to Question 1:

The type of delusion seen in this patient is a nihilistic delusion.

Nihilism refers to the delusion of the non-existence of self or parts of self. It also includes an attitude of rejection of established values or extreme skepticism regarding moral and value judgments. The patient may believe that they are going to die soon or their organs are rotting. Delusion of nihilism is often seen as a mood-congruent delusion in depression with psychotic symptoms. This patient has three major symptoms of depression — low energy, depressed mood, and lack of interest in the surroundings. However, these symptoms have to be present for a period of at least 14 days to make the diagnosis of major depressive disorder.

Other options:

Option A: Cotard's syndrome was first described by Jules Cotard. He described a psychotic disorder characterized by severe depression and nihilistic delusions regarding one's own body or existence. In the DSM or ICD, Cotard's syndrome is not classified separately. Nihilistic delusions have been mentioned as examples of mood-congruent delusions in a depressive episode with psychotic features.

Option B: Patients with hypochondriacal delusions are fixated on one or more somatic symptoms as evidence of some severe, yet undetected disease. They undergo repeated and unnecessary medical examinations and tests. There are frequent deteriorations of relationships with their doctors when nothing suggesting a disease is ever found, and this leads to doctor shopping.

Option D: Delusions are false and fixed ideas that are not shared by other people. Bizarre delusions are the ones whose content is out of the realm of possibility, for example, a patient believing that aliens abduct him and take him to Mars every night. Typically, patients with schizophrenia tend to have bizarre delusions.

Solution to Question 2:

La belle indifference is seen in conversion disorder.

La belle indifference refers to a lack of concern towards the symptoms, despite the apparent severity of the disability. Although not pathognomonic of conversion disorder, it is often seen in

association with it.

Conversion disorder is also known as functional neurological symptom disorder or dissociative disorder. It includes acute and temporary loss of motor or sensory function without any underlying neurological disease. Because the onset of such symptoms often coincides with psychological issues such as internal conflict, it was speculated that these issues converted to neurological symptoms. Conversion disorder can present with symptoms such as paresis, non-epileptic seizures (NES), blindness, and anesthesia of the extremities. One of the gait disturbances which can be seen is astasia-abasia — a wildly ataxic, staggering gait with gross and irregular truncal movements, and thrashing and waving arm movements. The patient often has a carefree attitude towards his illness as mentioned above.

Patients of conversion disorder achieve primary gain by keeping internal conflicts outside of their awareness. These conflicts manifest as physical symptoms. As a result of being sick, the patients are excused from obligations and receive support that might not otherwise have been available. This is referred to as secondary gain.

Solution to Question 3:

The SCOFF (Sick, Control, One, Fat, Food) questionnaire is a screening tool for eating disorders (anorexia nervosa and bulimia nervosa).

It consists of 5 questions addressing the core features of these disorders. The SCOFF questions are:

- Do you make yourself Sick because you feel uncomfortably full?
- Do you worry that you have lost Control over how much you eat?
- Have you recently lost more than One stone (~6.35 kg) in a 3-month period?
- Do you believe yourself to be Fat when others say you are too thin?
- Would you say that Food dominates your life?

A 'Yes' answer to 2 or more questions out of the 5 indicates likely anorexia or bulimia.

Solution to Question 4:

Formication, which refers to the sensation of insects crawling beneath the skin, is a feature of cocaine abuse. It is also referred to as Magnan's symptoms or cocaine bugs.

The symptoms of acute stimulant intoxication include agitation, irritability, aggression, and an increase in psychomotor activity. On examination, tachycardia, hypertension, and mydriasis may be seen. When psychotic symptoms such as delusions and hallucinations are present due to stimulants such as cocaine, it is referred to as stimulant-induced psychotic disorder. Paranoid delusions and hallucinations are hallmarks of stimulant-induced psychosis. Tactile hallucinations in the form of formication are seen in cocaine use. The presence of these symptoms depends on the dose and duration of the substance abuse and thus may not be seen in all patients. Cocaine-induced psychotic disorders are more frequently seen in intravenous users and those who

use crack (an extremely potent and addictive form of cocaine that can be smoked).

Other options:

Option A: Oneroid state is a dream-like state which can happen because of substance use or some organic disorder.

Option C: Dipsomania refers to a compulsive urge to drink alcoholic beverages. Patients suffering from alcohol-use disorders feel a need for daily use of large amounts of alcohol for adequate functioning throughout the day.

Option D: Carphologia (also known as floccillation) refers to the action of trying to pick imaginary objects. The patient will usually be trying to pick lint from his clothes or the bedsheet. This is often seen in delirium. Datura intake can lead to a delirious state and may show carphologia.

Solution to Question 5:

According to ICD-11, dependence on any substance is characterized by:

- Strong internal drive to use the substance, which the patient is unable to control.
- Increased priority is given to the use of the substance over other activities.
- Persistence of substance use despite harmful consequences.
- Presence of the subjective sensation of urge or craving to use the substance.
- Tolerance to the effect of the substance; the patient may require higher amounts of the substance to achieve the same effect.
- Appearance of withdrawal symptoms on cessation or reduction in the dose of the substance.

The diagnosis of dependence can be made if these symptoms are present for at least a period of 3 months.

Solution to Question 6:

In this scenario of a pregnant woman in her third trimester with bipolar disorder with multiple past episodes and currently stable on lithium, the best next step in management is to decrease lithium dosage within therapeutic levels.

Lithium is a mood stabilizer with a very narrow therapeutic index and is the first line of management in bipolar disorder. Lithium therapy should be continued throughout the pregnancy in women who have severe, frequent episodes, and these patients should be counseled about the reproductive risks associated with therapy. Ideally, the next best step in this scenario would be to monitor lithium blood levels monthly until 34 weeks, and weekly monitoring thereafter until delivery. However, dose reduction can be considered closer to delivery as there may be an increased risk of lithium toxicity in the fetus. According to NICE guidelines, it is recommended to stop lithium during active labour, measure lithium levels 12 hours after the last dose and reestablish prophylactic levels after delivery.

First and second trimester are characterised by a significant decrease of lithium blood levels with a risk of subtherapeutic levels. In third trimester and the postpartum, lithium levels gradually return to their preconception level which implicates that in this period clinicians need to be aware of the risk of lithium intoxication (Option B).

Bipolar disorder has been associated with adverse perinatal outcomes such as preterm birth. Medication discontinuation in pregnancy (Option A) can pose an increased risk for relapse. Moreover, symptomatic women during pregnancy more likely have postpartum relapses that can be severe.

Typical therapy for bipolar disorder includes mood stabilizers such as lithium, valproic acid, and carbamazepine and antipsychotic medications. Bipolar disorder during pregnancy is a complicated condition that is best treated in tandem with psychiatric care. Risks and advantages of mood stabilizers, some of which are teratogenic, must be considered while making decisions.

For example, lithium use during the first trimester has been linked to Ebstein anomaly, although very rarely. Fetal echocardiography is recommended for lithium-exposed fetuses. Fetal lithium exposure is also associated with neonatal goiter, CNS depression, hypotonia (floppy baby syndrome), and cardiac murmur.

Maternal adverse effects of lithium can be acute such as tremors, ataxia, weakness, polyuria, polydipsia or chronic such as nephrogenic diabetes insipidus, thyroid dysfunction, and hyperparathyroidism.

Due to all these adverse effects, patients on lithium therapy should have routine monitoring of serum electrolytes, serum calcium, thyroid profile, renal function tests, ECG, and serum lithium levels.

Other options :

Option D : It is not necessary to switch to valproate in the third trimester as the patient is already stable on lithium. Use of valproate in the first trimester is associated with significant teratogenicity.

Solution to Question 7:

HEADSSS assessment is done in adolescents to determine early symptoms of mental disorders and functional impairment in major life domains such as home and family, school, peers, and community.

These domains are included in the HEADSSS (Home, Education, Activities, Drugs, Sexuality, Suicide/Depression, Safety) Interview Guide. An adolescent's risk and protective factors are identified during each visit using this assessment. Risk factors increase the likelihood that an adolescent will engage in risky behaviors, whereas protective factors build an adolescent's resiliency and ability to cope with stress.

Examples of questions in HEADSSS assessment :

Home	Who lives at home, and where? Do you have your own room? What are relationships like at home? What do parents and relatives do for a living?
Education	School/grade performance-any recent changes? Any dramatic past changes? Any years repeated/classes failed? Suspension, termination, dropping out? Any current or past employment?
Activities	On own, with peers (what do you do for fun? where? when?) Hobbies-other activities? History of arrests-acting out-crime?
Drugs	Use by peers? Use by young person? (include tobacco, alcohol) Use by family members? Amounts, source, frequency, patterns of use/abuse
Sexuality	Degree and types of sexual experience and acts? Number of partners? Sexually transmitted diseases-knowledge and prevention? Contraception? Frequency of use? History of pregnancy/abortion?
Suicide / Depression / Self Image	Sleep disorders, appetite/eating behavior changes, emotional outbursts and highly impulsive behavior, history of withdrawal/isolation, hopeless/helpless feelings, suicidal ideation History of past suicide attempts, depression, psychological counseling
Safety	Do you ever feel unsafe?

Psychiatry INI-CET 2024

Question 1:

Which of the following is not a priority condition under the WHO Mental health gap action plan?

- a) Schizophrenia and other psychotic disorders
- b) Obsessive compulsive disorder
- c) Depression
- d) Mental health disorders in children

Question 2:

Which of the following is not true about 'rapid cyclers'?

- a) Occurs commonly in men
- b) Commonly seen with concomitant hypothyroidism
- c) At least 4 distinct episodes per year
- d) Can be triggered by antidepressants

Question 3:

Which among the following is not part of the Beck's cognitive theory on depression?

- a) Cognitive decline
- b) Automatic thoughts
- c) Dysfunctional beliefs
- d) Introjection

Question 4:

Which of these is not an essential feature for the diagnosis of autism spectrum disorder?

- a) Persistent deficits in social communication
- b) Abnormalities in socializing

- c) Restrictive patterns of behaviour and interests
- d) Cognitive impairment

Question 5:

All the following can be treated with supportive psychotherapy except

- a) Person with cognitive decline and physical symptoms
- b) Person who has good motivation and control over frustration
- c) Patient with acute psychiatric illness along with crisis
- d) Patient with severe chronic disease and ego dysfunctioning

Answer Key

Question No.	Correct Option
1	b
2	a
3	d
4	d
5	b

Detailed Explanations

Solution to Question 1:

Obsessive compulsive disorder is not a priority condition under WHO Mental health gap action plan.

WHO aims to offer health planners, policymakers, and donors a comprehensive and cohesive set of initiatives and projects to enhance the provision of care for mental, neurological, and substance use disorders through the Mental Health Gap Action Programme (mhGAP).

The priority conditions identified by the above criteria for mhGAP are depression, schizophrenia and other psychotic disorders, suicide, epilepsy, dementia, disorders due to use of alcohol, disorders due to use of illicit drugs, and mental disorders in children.

The intervention package includes evidence-based strategies for prevention and management, considering cost-effectiveness, severity of disorders, poverty reduction, and human rights protection.

Solution to Question 2:

Option A is the false statement. Rapid cycling disorder is more common in females.

Rapid cycling disorder is diagnosed when the patient has a combination of 4 or more bipolar episodes within the last 12 months. Two separate episodes can be demarcated from one another by either switch in polarity or full/partial remission for at least 2 months. Manic and hypomanic episodes are considered to be on the same pole, hence a switch in polarity includes an episode of mania/hypomania with an episode of depression.

Possible risk factors for rapid cycling disorder include female patients, diabetes, childhood abuse or family trauma, and alcoholism or substance abuse. Rapid-cycling bipolar patients are significantly more likely than non-rapid-cycling patients to have a lifetime history of comorbidities like anxiety disorders, ADHD, OCD, bulimia nervosa, and substance use disorders.

The drug of choice for prophylaxis of rapid cycling disorder is valproate

Other options

Option B: Conditions like hypothyroidism and diabetes are possibly involved in some cases of rapid cycling .

Option D: Usage of antidepressants could worsen rapid cycling in bipolar depression.

Solution to Question 3:

Introjection is not a part of the Beck's cognitive theory of depression.

Aaron Beck's cognitive triad consists of:

- Beliefs about oneself - a pessimistic self-image
- Perceptions of the surroundings - a tendency to see the world as unfriendly and challenging,
- Thoughts about the future - anticipating pain and defeat.

The psychodynamic understanding of depression, initially defined by Sigmund Freud and later expanded by Karl Abraham, is considered the classic view of depression. This theory includes four main points: (1) disturbances in the infant-mother relationship during the oral phase (the first 10 to 18 months of life) create a predisposition to depression later in life; (2) depression can arise from the real or imagined loss of an important object; (3) introjection of the lost object serves as a defense mechanism to cope with the distress of the loss; and (4) because the lost object is associated with both love and hate, feelings of anger are turned inward toward the self.

Solution to Question 4:

Cognitive impairment does not form part of the diagnostic criteria for autism spectrum disorder.

The DSM-5 diagnostic criteria for autism spectrum disorder are as follows:

- Persistent deficits in social communication and social interaction across multiple contexts, currently or by history
- Restricted, repetitive patterns of behavior, interests, or activities, currently or by history
- Symptoms must be present in the early developmental period (but may not become fully manifest until social demands exceed limited capacities, or may be masked by learned strategies in later life).
- Symptoms cause clinically significant impairment in social, occupational, or other important areas of current functioning.
- These disturbances are not better explained by intellectual disability or global developmental delay. Intellectual disability and autism spectrum disorder frequently co-occur; to make comorbid diagnoses of autism spectrum disorder and intellectual disability, social communication should be below that expected for the general developmental level.

Solution to Question 5:

Supportive psychotherapy is not indicated in a person who has good motivation and control over frustration. Such individuals are better treated by expressive psychotherapy.

Supportive psychotherapy places major etiological emphasis on external rather than intrapsychic events, particularly on stressful environmental and interpersonal influences on a severely damaged self. The general aim of supportive treatment is the amelioration or relief of symptoms through behavioral or environmental restructuring within the existing psychic framework. This often means helping the patient to adapt better to problems and to live more comfortably with his or her psychopathology.

Supportive therapy uses several methods, either singly or in combination, including warm, friendly, strong leadership; help in developing pleasurable activities (e.g., hobbies); removal of excessive strain, when possible; medication to alleviate symptoms; and guidance and advice in dealing with current issues. This therapy uses techniques to help patients feel secure, accepted, encouraged, and not anxious.

Supportive psychotherapy is generally indicated for those patients for whom classic psychoanalysis or insight-oriented psychoanalytic psychotherapy is typically contraindicated—those who have poor ego strength and whose potential for decompensation is high.

Amenable patients fall into the following major areas:

- Individuals in acute crisis or a temporary state of disorganization and inability to cope, whose intolerable life circumstances have produced extreme anxiety or sudden turmoil (e.g., individuals going through grief reactions, illness, divorce, job loss, or who were victims of crime, abuse, natural disaster, or accident);
- Patients with chronic severe pathology with fragile or deficient ego functioning (e.g., those with latent psychosis, impulse disorder, or severe character disturbance);
- Patients whose cognitive deficits and physical symptoms make them particularly vulnerable and, thus, unsuitable for an insight-oriented approach (e.g., certain psychosomatic or medically ill persons);

- Individuals who are psychologically unmotivated (e.g., patients who come to treatment in response to family or agency pressure and are interested only in immediate relief or those who need assistance in very specific problem areas of social adjustment as a possible prelude to more exploratory work).

Because support forms a tacit part of every therapeutic modality, it is rarely contraindicated as such. The typical attitude regards better-functioning patients as unsuitable not because they will be harmed by a supportive approach, but because they will not be sufficiently benefited by it.

Psychiatry NEET 2024

Question 1:

A patient comes with complaints of difficulty falling asleep. He consumes black coffee one hour before bed. Which of the following is the role of caffeine in the wakefulness?

- a) Caffeine has no effect, and he can continue taking black coffee
- b) Caffeine causes release of chemical from locus coeruleus that causes awakening
- c) Caffeine has histaminic action in brain that leads to awakening
- d) Caffeine inhibits adenosine receptors, thus leading to awakening

Question 2:

Which of the following SNRI is used in the treatment of ADHD?

- a) Guanfacine
- b) Methylphenidate
- c) Reboxetine
- d) Modafinil

Question 3:

A 45-year-old woman presents with a feeling of nervousness and has abdominal pain and diarrhoea. She has had a history of heartburn, and indigestion for the past 25 years. She gives similar complaints in the family of being "tense and nervous". What other symptoms might you expect to see in this condition?

- a) Tingling of extremities
- b) Neologism
- c) Idea of reference
- d) Hallucination

Question 4:

A girl was referred to the psychiatry department by the plastic surgeon. She had visited multiple plastic surgery departments requesting full facial reconstruction. She complains of dark blemishes and spots on her face, which are not present. She always says that she is ugly and looks constantly in the mirror and the back of the spoon and windows. Her mother assured her of no such issues, but still, she persists in getting the surgery. What should the psychiatrist do?

- a) Start on atypical antipsychotics
- b) Give fitness for surgery
- c) Start her on selective serotonin reuptake inhibitors
- d) Initiate insight-oriented psychotherapy

Question 5:

A girl presented with multiple episodes of loss of consciousness. The episodes usually happen in the presence of the family members in the morning. She regains consciousness after 15 to 30 minutes and usually remembers the events during these episodes. There is no history of tongue bite or urinary incontinence. What should be done in this scenario?

- a) Start behavioural therapy
- b) Initiate aversion therapy
- c) Give valproate
- d) Start on ketogenic diet

Question 6:

If a child with ADHD goes untreated into adolescence, what is he most likely to develop?

- a) Selective mutism
- b) Separation anxiety disorder
- c) Conduct disorder
- d) Binge eating disorder

Question 7:

A chronic alcoholic met with a vehicular accident. His blood alcohol level is >300 mg/dl. He needs hospitalization and will be withheld from taking alcohol. The drug that will be used to manage his withdrawal symptoms is:

- a) Methyl alcohol
- b) Fomepizole
- c) Lorazepam
- d) Buspirone

Question 8:

A 19-year-old girl child came agitated to the emergency department after consuming some drug with a previous history of suicidal tendencies. The parents do not know the mode and route of the drug administered. She presented with tachycardia, tachypnea, diaphoresis, and dilated pupils. What is the likely drug administered?

- a) Chlorpheniramine
- b) Morphine
- c) Heroin
- d) Cocaine

Question 9:

A 19-year-old presented to the psychiatry OPD. He has a charming persona and is well-composed and behaved in front of interviewers. His mother complains of him getting into fights and bunking classes, drinking alcohol and getting involved in police cases. What kind of personality disorder does he have?

- a) Paranoid
- b) Antisocial
- c) Narcissistic
- d) Schizotypal

Answer Key

Question No.	Correct Option
1	d
2	c
3	a

4	c
5	a
6	c
7	c
8	d
9	b

Detailed Explanations

Solution to Question 1:

The role of caffeine in the wakefulness in this patient is that caffeine inhibits adenosine receptors, thus leading to awakening.

Caffeine, a commonly used stimulant, plays a significant role in promoting wakefulness and alertness. It functions primarily as an antagonist at adenosine A_{2A} receptors in the brain. Adenosine, a natural neuromodulator which accumulates in the basal forebrain area promotes sleepiness and relaxation by activating these receptors. By blocking adenosine A_{2A} receptors, caffeine counteracts these sedative effects, thus helping to increase wakefulness and reduce fatigue.

In addition to its receptor antagonism, caffeine inhibits cyclic nucleotide phosphodiesterases, which leads to increased levels of cyclic AMP (cAMP) in the brain. This action enhances the release of neurotransmitters such as norepinephrine (NE) and dopamine (DA), further contributing to increased neural activity and alertness. Caffeine is rapidly absorbed from the digestive tract and crosses the blood-brain barrier easily, affecting brain function soon after consumption.

Caffeine's stimulating effects can also be influenced by factors such as metabolism, which varies widely among individuals. For instance, smoking can reduce caffeine's half-life, making it clear from the body faster, while oral contraceptives can prolong its effects. Therefore, consuming caffeine close to bedtime, as in the case of drinking black coffee an hour before bed, can interfere with sleep by sustaining wakefulness and reducing the ability to fall asleep.

Solution to Question 2:

The SNRI that can be used for the treatment of ADHD is Reboxetine.

Reboxetine, a SNRI has shown potential efficacy in treating ADHD, particularly in patients who do not respond to methylphenidate. Common side effects of reboxetine include drowsiness, sedation, and gastrointestinal issues.

In the treatment of ADHD, pharmacotherapy is the primary approach, with central nervous system (CNS) stimulants being the most effective. Stimulants such as methylphenidate and amphetamine salts are commonly used due to their high efficacy in managing symptoms,

although they come with risks of dependence and addiction. For patients who cannot tolerate stimulants or have a family history of addiction, non-stimulant options are preferred. These include atomoxetine, a norepinephrine reuptake inhibitor (NRI), and guanfacine, an alpha-2 adrenergic agonist.

Modafinil is used primarily for narcolepsy and is not a standard treatment for ADHD.

Solution to Question 3:

In this scenario, given the patient's feelings of nervousness, abdominal pain, diarrhoea, and a history of long-term digestive issues along with familial history, you might also expect to see tingling of extremities as a symptom of Generalized Anxiety Disorder (GAD).

According to DSM-5 (and ICD-11) criteria, GAD is characterized by excessive and uncontrollable anxiety and worry about various aspects of life, persisting for at least six months. This anxiety is typically accompanied by at least three additional symptoms from a specific list, which includes:

- The presence of excessive anxiety and worry about a variety of topics, events, or activities.
- Worry occurs more often than not for at least six months and is clearly excessive.
- The worry is experienced as very challenging to control.
- The worry in both adults and children may easily shift from one topic to another.
- The anxiety and worry are accompanied by at least three of the following physical or cognitive symptoms (In children, only one of these symptoms is necessary for a diagnosis of GAD):
- Edginess or restlessness
- Tiring easily; more fatigued than usual impaired concentration or feeling as though the mind goes blank
- Irritability (which may or may not be observable to others)
- Increased muscle aches or soreness
- Difficulty sleeping (due to trouble falling asleep or staying asleep, restlessness at night, or unsatisfying sleep)

Neologism, idea of reference, and hallucinations are not typical symptoms of GAD but are more associated with other psychiatric conditions such as psychotic disorders like schizophrenia.

Solution to Question 4:

The given history suggests the presence of body dysmorphic disorder (BDD) in the patient, as she is preoccupied with her appearance and self-perceived flaws, despite being told otherwise by her family. The psychiatrist should start her on selective serotonin reuptake inhibitors(SSRI).

BDD is characterized by persistent preoccupations about one or more perceived defects or flaws in one's appearance, but the defects or flaws are negligible or not even observable to others. In response to their appearance concerns, individuals with BDD demonstrate a range of mental acts:

- Comparing themselves with others
- Checking in the mirror
- Camouflaging their perceived flaws
- Avoidance- from minor social avoidance to being housebound.

The specific focus of BDD on appearance serves to differentiate it from other conditions characterized by repetitive behaviours.

The treatment for BDD typically involves a combination of SSRIs and psychotherapy. SSRIs, such as fluoxetine and clomipramine, have been shown to reduce symptoms in many patients, helping alleviate the obsessive thoughts and anxiety associated with BDD.

Psychotherapy, particularly cognitive-behavioural therapy (CBT), is crucial as it addresses the distorted self-image and underlying emotional issues. CBT helps patients reframe their thoughts about appearance and develop healthier coping strategies.

It's important to note that while surgical interventions may be sought by patients, they are often ineffective and can lead to further dissatisfaction and distress. Therefore, focusing on mental health treatment is essential. Regular follow-ups and a supportive therapeutic relationship can significantly enhance the effectiveness of treatment.

Other options:

Option A: Start on atypical antipsychotics: Antipsychotics are generally not the first-line treatment for BDD. They may be considered in cases with comorbid psychotic symptoms, but this patient does not exhibit hallucinations or delusions.

Option B: Give fitness for surgery: Giving fitness for surgery would be inappropriate and potentially harmful. Surgery is not the solution for BDD, as patients typically remain dissatisfied with the results and continue seeking more surgeries.

Option D: Initiate insight-oriented psychotherapy: While psychotherapy, particularly cognitive-behavioral therapy (CBT), is effective for BDD, insight-oriented psychotherapy (focused on understanding unconscious conflicts) is less effective than targeted approaches like CBT in managing this condition.

Solution to Question 5:

In this scenario, the girl is experiencing episodes of loss of consciousness that may suggest conversion disorder. The most appropriate management is to start behavioural therapy to help address her symptoms and underlying stressors.

Conversion disorder, also referred to as functional neurological symptom disorder in the DSM-5, involves symptoms or deficits that affect voluntary motor or sensory functions, suggesting another medical condition. However, these symptoms are believed to arise from psychological factors, often triggered by conflicts or stressors. The symptoms are not intentionally produced, are not the result of substance use, and are not limited to pain or sexual issues. The primary gain associated with conversion disorder is psychological rather than social, monetary, or legal.

Common presenting symptoms include:

- Non-epileptic seizures
- Weakness and/or paralysis
- Movement disorders
- Speech or visual impairment
- Swallowing difficulty
- Sensory disturbances
- Cognitive symptoms

Resolution of conversion disorder symptoms often occurs spontaneously but can be enhanced through supportive or behavioural therapy. For patients who are resistant to psychotherapy, it can be beneficial to focus on stress and coping strategies rather than directly addressing their symptoms as imaginary, as this may exacerbate their condition.

Additional techniques, such as hypnosis, anxiolytics, and behavioural relaxation exercises, may also be effective in managing symptoms. In certain cases, medications like parenteral amobarbital or lorazepam can help gather further information, especially if the patient has experienced a recent traumatic event.

Psychodynamic approaches, including psychoanalysis and insight-oriented psychotherapy, allow patients to explore their internal conflicts and the symbolic meanings of their symptoms.

Solution to Question 6:

If a child with ADHD goes untreated into adolescence, among the following options he is most likely to develop conduct disorder.

The course of ADHD is variable. The majority of the cases persist into adult life. The hyperactivity may sometimes disappear, but the decreased attention span and impulse-control problems usually persist. Patients with ADHD are vulnerable to conduct disorder, antisocial behaviour, substance use disorders, and mood disorders.

ADHD is a neuro psychiatric condition characterized by a pattern of diminished attention span and increased impulsivity or hyperactivity. The prefrontal cortex of the brain has been implicated in its etiology and dopamine is the main neurotransmitter.

ADHD can begin during infancy but is usually recognized between 1-3 years of age. ADHD was known as minimal brain dysfunction earlier. The characteristics of children with ADHD, in order of frequency, are

- Hyperactivity
- Attention deficit (short attention span, distractibility)
- Impulsivity (action before thought, abrupt shifts in activity, lack of organization)
- Memory and thinking deficits
- Specific learning disabilities
- Speech and hearing deficits.

According to DSM-5 criteria, a diagnosis of ADHD requires symptoms to be present before age 12, persist for at least six months, negatively impact social, academic, or occupational activities, and occur in two or more settings. Pharmacotherapy is the first-line treatment, with CNS stimulants like methylphenidate and amphetamine salts being the most effective due to their mild side effects. Non-stimulant options include atomoxetine, clonidine, and guanfacine. Additionally, psychosocial interventions—such as academic organization skills, parent training, behaviour modification, cognitive-behavioural therapy (CBT), and social skills training—are used as adjuncts to medication.

Note: Conduct disorder is characterized by a persistent pattern of behaviour that violates the basic rights of others and disregards societal rules. The severity of the behaviours can lead to emergencies, especially if they are alarming to parents. Individuals with severe and chronic symptoms may later be diagnosed with antisocial personality disorder after age 18. Treatment typically involves behaviour modification strategies and, if necessary, placement in a corrective institution.

Other options:

Option A: Selective mutism refers to the lack of speaking (being mute) in one or more specific social situations, commonly in school, but speaking well in other situations, commonly at home. The child may be either completely silent at school or maybe just whispering. Rarely, a child may be silent at home but communicate normally at school.

Option B: Separation anxiety disorder is characterized by developmentally inappropriate and excessive anxiety when the patient is separated from their attachment figure. It is a normal developmental phenomenon in infants. It peaks between 9 months and 18 months and usually diminishes by two and a half years of age.

Option D: Binge-eating disorder is the most common eating disorder. It is characterized by rapid intake of abnormally large quantities of food over a short period of time, even when not hungry. There are feelings of guilt about the eating episodes, but these patients do not compensate in any way following the episodes of binge.

Solution to Question 7:

The first-line drug used in the management of this patient's withdrawal symptoms is intravenous(IV) lorazepam.

Delirium tremens is the most severe form of alcohol withdrawal syndrome. It occurs usually within 3–4 days of alcohol withdrawal. Acute infection, hepatitis and pancreatitis predispose delirium tremens.

Clinically, patients present with confusion, agitation, and fluctuating levels of consciousness. Visual or tactile hallucinations along with features of autonomic hyperactivity like tachycardia, diaphoresis, fever, insomnia, and hypertension are also seen.

Delirium tremens is treated using benzodiazepines, starting with a higher dosage, which is gradually tapered as withdrawal symptoms disappear. Benzodiazepines such as IV lorazepam or oral chlordiazepoxide (every 4 hours) are given.

In the case of benzodiazepine-resistant cases, IV phenobarbital is the first-line drug, and IV propofol is the second-line drug.

It is also recommended to administer thiamine in patients with alcohol-related disorders to prevent Wernicke-Korsakoff syndrome

Solution to Question 8:

In the given clinical scenario, the patient presenting with tachycardia, tachypnea, diaphoresis, and dilated pupils after consuming some drug with a previous history of suicidal tendencies points towards a diagnosis of cocaine toxicity leading to sympathetic toxidrome.

Cocaine is a CNS stimulant obtained from coca plants belonging to the family Erythroxylaceae. The route of intake is by snorting, IV route and smoking. Tattoos may be made to cover the injection marks.

It inhibits the uptake of dopamine, norepinephrine, and serotonin.

At lower doses, it leads to alertness but at higher doses, it produces stereotyped behavior, involuntary actions, and paranoia. There is a transient euphoria. Tactile hallucinations (Magan's symptom or cocaine bugs) are characteristic.

There is a dose-dependent increase in heart rate and blood pressure. It causes dilation of pupils and is associated with chills and perspiration. In severe cases, intracranial bleeding due to increased blood pressure, and myocardial infarction due to coronary vasospasm can occur. Ultimately, the patient goes into a stage of depression with loss of reflexes, paralysis of muscles, cardiorespiratory failure, and coma.

Management - Administration of activated charcoal and gastric lavage with warm water and potassium permanganate in case of ingestion. Benzodiazepines to control heart rate/seizures. Amyl nitrite is the antidote given via inhalation. Thiamine 100 mg IV and Naloxone 2 mg IV can also be given.

Other options:

Option A: Chlorpheniramine is an effective H₁ antagonist that tends to be mildly sedative, making it suitable for daytime use. A notable number of patients experience side effects involving central nervous system stimulation.

Option B and C: Morphine and Heroin are isolated from opium. The clinical features include a euphoric feeling in the stage of excitement followed by headache, giddiness, and a tendency to sleep in the stage of stupor. Intoxication is characterized by bradycardia, hypotension, respiratory depression, hypothermia, and pinpoint pupils. Respiratory depression can even result in death.

The image below shows the Erythroxylum plant:



Solution to Question 9:

The patient most likely has antisocial personality disorder. Although the patient seems charming and well-composed, he is aggressive and clearly exhibits disregard for social norms, and shows no remorse for his actions. These findings are characteristic of antisocial personality disorder.

Antisocial Personality Disorder is characterized by a long-term pattern of manipulation, exploitation, or violation of others' rights. Individuals with antisocial personality disorder tend to show no regard for right or wrong and often disregard the feelings, rights, and safety of others.

Other options:

Option A: Paranoid personality disorder is characterized by unwarranted suspicion, hypersensitivity, jealousy, envy, rigidity, excessive self-importance, and a tendency to blame and ascribe evil motives to others.

Option C: Narcissistic personality disorder is marked by a heightened sense of self-importance. Individuals with this personality disorder display a lack of empathy and grandiose feelings. This is to mask their fragile self-esteem and vulnerability.

Option D: Individuals with schizotypal personality disorder appear strange to surrounding people. They have ideas of reference, odd or magical thinking, inappropriate or constricted affect, and suspiciousness. It is most commonly associated with depression in nearly half of the patients.